

Patently Peculiar: Patents and Innovation in the United Kingdom of the Netherlands*

Homer Wagenaar[†]

Christopher L. Colvin[‡]

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Abstract

We study the patent system of the United Kingdom of the Netherlands (1815–1830), a state that combined an early-industrialising region corresponding to present-day Belgium and a late-industrialising region corresponding to what is now the Netherlands. The system distinguished formally between patents of invention and patents of importation, protecting both domestic inventions and the introduction of foreign technologies. It also granted administrators considerable discretion over patent fees, terms and conditions. Using a new hand-collected dataset covering all patent applications, including granted, withdrawn and rejected cases, we analyse application outcomes across sectors, regions and patent types. We show that administrators systematically differentiated between patents of invention and importation in ways consistent with a technology policy that encouraged the domestic introduction of useful foreign technologies. We also find that administrative decisions reflected assessments of novelty, utility and applicant characteristics, without evidence of systematic discrimination. Despite high fees, the system remained broadly accessible across social classes. Patents of invention were more commonly associated with technically skilled inventors, whereas patents of importation were more often sought by commercially oriented applicants. Overall, the regime combined accessibility with effective screening and illustrates how discretionary patent administration could support technology transfer during a formative period of industrialisation.

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[†] Belgian Competition Authority, Brussels.

[‡] Department of Economics, Queen’s University Belfast, and Centre for Economics, Policy and History. Corresponding author. Email: chris.colvin@qub.ac.uk.

We conduct a quantitative analysis of a new hand-collected dataset of *all* patent applications – granted, rejected, and withdrawn – submitted to the government of the United Kingdom of the Netherlands (UKNL), a sovereign state broadly corresponding to the territories of present-day Belgium and the Netherlands and in personal union with Luxembourg.¹ Established in 1815 at the conclusion of the Napoleonic Wars as a buffer against future French aggression, the UKNL introduced a unified patent system in 1817, centrally administered by a single bureaucracy from its alternating capitals of Brussels and The Hague.² This patent system remained in place even after the state’s *de facto* dissolution in 1830; the newly independent Kingdom of Belgium retained and strengthened the 1817 patent law, while it gradually fell into disuse in the rump Kingdom of the Netherlands, culminating in its abolition there in 1869.³ We examine the use of this patent system during a critical period of industrialisation in the southern half of the UKNL state, the region that became the world’s second industrial nation.⁴

The UKNL’s new patent system was rather peculiar. It was principally based on France’s 1791 patent law, but it augmented French legal practice by giving civil servants significant discretion over its administration.⁵ Applicants could apply for five-, ten-, or fifteen-year terms, had to pay high upfront fees but no renewal fees, and were required to put inventions into production within two years. Notably, administrators adopted some informal French practices, such as substantive examination and fee discounts for poorer applicants.⁶ However, unlike France, the UKNL state empowered its civil servants to reject applications, adjust patent terms, introduce conditions, and change fees on a case-by-case basis following a ‘wholistic’ assessment of novelty, utility, and financial means, thus introducing a discretionary ‘Dutch dimension’ into patent governance.⁷ Somewhat strangely from a modern perspective, patent specifications were to be kept secret until the patents expired, conferring benefits normally exclusive to trade secrecy.⁸

¹ Willem Frederik, Prins van Oranje-Nassau, proclaimed himself King William I of the Netherlands on 16 March 1815 and became Grand Duke of Luxembourg on 9 June of the same year, and so the UKNL and Luxembourg were in personal union (Kennedy, *A concise history*). Both polities were *de facto* administered as a single jurisdiction for the purposes of the patent system.

² UKNL patent law did not apply to the UKNL’s colonial possessions in Africa, the Americas, and Southeast Asia. Nor did it apply to Neutral Moresnet, the *de facto* condominium governed through an often improvised and only loosely coordinated arrangement between the UKNL and Prussia (Press, *Europe’s Little Anarchy*, p. 105).

³ Van Gompel, ‘Patent abolition’; Wagenaar, ‘The patent system’.

⁴ Mokyr, *Industrialization*; Philips and Buyst, ‘Regional industrialization’.

⁵ Wagenaar, ‘The patent system’.

⁶ See Baudry, ‘Une histoire’ (pp. 226–29), and Baudry, ‘Examining inventions’, for a description of these informal practices in the French patent system; and Nuvolari, Tortorici and Vasta, ‘British-French technology’ for a cliometric treatment of the French system.

⁷ This discretionary practice was a continuation of an earlier Dutch tradition of patent-system administration dating from the Dutch Republic (described in Davids, ‘Patents and patentees’).

⁸ Some of these features made UKNL patent law and practice closer to the contemporary patent systems of German states. See Donges and Selgert, ‘Do legal differences matter?’; Donges and Selgert ‘Technology transfer’; and Lehmann-Hasemeyer and Streb, ‘Discrimination against foreigners’.

One notable feature of the UKNL patent system was the formal separation between patents of invention and patents of importation (or introduction). Unlike patent systems that require global novelty, the UKNL defined novelty in relation to the domestic economy, granting patents for both original inventions and the introduction of technologies developed abroad.⁹ Although both types of patents conferred exclusive rights, patents of importation were typically subject to distinct conditions, including shorter terms linked to the original foreign patent and stricter working requirements. This dual structure reflects the earlier tradition of using patent systems not only to reward invention, but also as policy instruments to promote industrial catch-up and technological diffusion.¹⁰

Elsewhere, Homer Wagenaar traces patent applications of all types through the UKNL's nascent patent system to show how its administrators implemented the new patent law and developed regularised informal practices.¹¹ He tracks the day-to-day operation of the patent system over time to see how patent law and practice changed in each successor state following Belgium's *de facto* independence in 1830. Wagenaar's institutional-legal history highlights the use of discretion in applying patent law, in contrast to the one-size-fits-all approach that is the norm in the design of intellectual property rights today.¹² We build on Wagenaar's earlier work with a data-driven exercise – the first to exploit the full archive of all UKNL patent applications.

Gerard Doorman remains the only prior scholar to collect and describe actual patent data for the UKNL, though his presentation is essentially limited to a few tables of aggregate patent counts.¹³ The most influential work that uses Doorman's patent data is by Joel Mokyr, who presents patent statistics to illustrate that the Southern Netherlands (i.e., present-day Belgium) industrialised earlier than the Northern Netherlands (i.e., present-day the Netherlands).¹⁴ An interesting fact highlighted by Mokyr's work is that both parts of the kingdom shared the same

⁹ This contrasts with the priority system under modern international patent law first established by the 1883 Paris Convention, which requires that the applicant be the first and original inventor (or successor in title), and under which the filing date of the first (priority) application establishes a limited 12-month window for any subsequent applications for the same invention in other patent systems (Lundstedt, 'The law').

¹⁰ Patents of importation are a continuation of an early modern practice of technology transfer. In England, patents for introducing foreign technologies were granted well before patents for original inventions became common (Bottomley, *The British patent system*). The concept became embedded in statutory patent regimes of countries that followed the French patent law of 1791, including Spain (Sáiz, 'Did patents').

¹¹ Wagenaar, 'The patent system'.

¹² Scotchmer, *Innovation and incentives*; Burk and Lemley, *The patent crisis*.

¹³ Doorman, *Het Nederlandsch octrooiwezen*.

¹⁴ Building on Mokyr, 'The Industrial Revolution in the Low Countries', and Mokyr, *Industrialization*, Richard Griffiths argued that the Netherlands suffered an 'industrial retardation', industrialising only very late in the nineteenth century (Griffiths, *Industrial retardation*). Griffiths's late dating has since been pushed back somewhat (van Zanden and van Riel, *The strictures*). The timing and geography of Belgium's early industrialisation have also been revised (Philips and Buyst, 'Regional industrialization').

patent institutions, but nevertheless had very different experiences of innovation and technical change.¹⁵

Other works that examine intellectual property rights in the Low Countries do not cover the early nineteenth century. For example, doctoral dissertations by De Favereau and Péters describe only the patent system of post-independence Belgium.¹⁶ And while the post-partition Dutch patent system has garnered more attention than its Belgian counterpart, this is always in the context of its abolition and the country's subsequent 'patentless period' (1869–1912).¹⁷ We fill a clear gap in the literature with a comprehensive statistical study of an underexplored period, adding especially to Mokyr's analysis by clarifying the pattern of industrialisation across the Low Countries, and revisiting the conclusion that having a patent system is not a sufficient condition for the Industrial Revolution to take root. Our focus on comparing patents of invention and patents of importation also allows us to analyse how formal institutions influenced the cross-border transfer of technology.

We explore how accessible the UKNL's patent system was, how it compared with contemporaneous systems elsewhere in Europe, and how application outcomes varied across several dimensions. Our dataset includes granted, rejected, and withdrawn applications – unlike other cliometric patent studies we are aware of, which focus only on granted patents rather than all patent applications.¹⁸ The one exception to this is new research by Stephen Billington and Joe Lane, who constructed a dataset of unsuccessful English patent applications between 1783 and 1834.¹⁹ As with Billington and Lane's study, our comprehensive dataset permits us to explore the correlates of a patent application's success or failure, as well as the conditions under which a patent is granted. Data were compiled from handwritten government archives and include inventor characteristics (occupation, location), which we have coded using the HISCO-HISCLASS scheme, and patent content, which we have classified into technology classes using

¹⁵ This relates to Mokyr's subsequent work on patents during Britain's Industrial Revolution, where he argues that they were neither a necessary nor a dominant driver of innovation, but functioned as one incentive among many within a broader institutional and cultural environment (Mokyr, 'Intellectual property rights'; Mokyr, *The enlightened economy*). Mokyr himself later shifted the emphasis towards the institutions underpinning the creation and diffusion of useful knowledge as an explanation for the Netherlands' comparatively late industrialisation (Mokyr, 'The Industrial Revolution and the Netherlands').

¹⁶ De Favereau, 'Faire germer le progrès', and Péters, 'Course à l'innovation'.

¹⁷ Doorman, *Het Nederlandsch octrooiwezen*; Machlup and Penrose, 'The patent controversy'; Schiff, *Industrialization*; den Hertog, 'De anti-octrooibeweging'; Gerzon, *Nederland*; Stokvis, 'Industrialisering'; Moser, 'How do patent laws'; Boldrin and Levine, *Against intellectual monopoly*; de Ridder, 'Octrooien onder vuur!'; and, most recently, van Gompel, 'Patent abolition'.

¹⁸ Andrews, 'Historical patent data'.

¹⁹ Billington and Lane, 'Success and failure', who show that private opposition acted as a *de facto* screening mechanism that shaped patenting behaviour and outcomes in England.

a machine-learning algorithm.²⁰ We investigate differences in treatment between domestic and foreign applicants, and between inventors from northern and southern provinces.

We find that administrative discretion did not produce unfair, arbitrary, or inconsistent outcomes for patent applicants, but instead functioned as a screening mechanism that calibrated patent conditions to balance the interests of inventors with broader government objectives. Despite high fees, the system was accessible to non-elite inventors, with similar social-class representation across regions. Demand for patent protection was driven by skill and market orientation. The predominant narrative in the historiography is that the UKNL's peculiar patent regime was destined to fail.²¹ By contrast, we show it functioned as what Douglass North and his co-authors would describe as an 'open access order institution', which was developing clear and well-understood rules and practices.²² Relatedly, we argue that the UKNL patent system constituted what Daron Acemoglu and James Robinson would describe as an 'inclusive economic institution', open to all irrespective of ability to pay.²³ Discretion was used consistently and constructively, challenging the idea that discretion necessarily leads to biased or time-inconsistent policy decisions, as Finn E. Kydland and Edward C. Prescott have argued.²⁴

Our study joins a growing body of cliometric work on patent systems in industrialising Europe.²⁵ This literature, in turn, builds on pioneering US studies.²⁶ Not only do we contribute by adding a new country case study, but our comparisons of findings across these studies offer new insights for scholars seeking to understand the state-of-the-art in this field. We demonstrate that outcomes under the UKNL system, despite its apparently peculiar design, were not wholly

²⁰ See van Leeuwen, Maas and Miles, *HISCO*, and van Leeuwen and Maas, *HISCLASS*, for the HISCO-HISCLASS scheme. See Billington and Hanna, 'That's classified!', for the patent classification algorithm.

²¹ Schiff, *Industrialization*, typifies the argument that the Dutch patent system was 'doomed' from the outset.

²² North, Wallis and Weingast, *Violence*, define open access orders as societies in which political and economic organisation is open, impersonal, and governed by general rules rather than privilege.

²³ Acemoglu and Robinson, *Why nations fail*, define inclusive economic institutions as those which create broad-based incentives and opportunities by securing property rights, enforcing the rule of law, and allowing widespread participation in economic and political life.

²⁴ Kydland and Prescott, 'Rules rather than discretion', use patent policy as an example of how rules can overcome the time-inconsistency problem in rewarding invention (p. 477).

²⁵ For England, Ireland and Scotland, see Bottomley, *The British patent system*, Billington, 'What explains patenting', and Billington and Lane, 'Procuring promising provisions'; for Spain, see Sáiz, 'Did patents of introduction'; for pre- and post-unification German states, see Streb, Baten and Yin, 'Technological and geographical knowledge', Donges and Selgert, 'Do legal differences', Donges and Selgert, 'Technology transfer', Donges and Selgert, 'The social background', and Lehmann-Hasemeyer and Streb, 'Discrimination'; for pre-unification Italy, see Nuvolari and Vasta, 'Independent invention'; for France, see Galvez-Behar, 'The patent system', and Nuvolari, Tortorici and Vasta, 'British-French technology'; and for the Nordic countries, see Andersson and La Mela, 'Nordic networks', Andersson and Tell, 'From fighting monopolies', and Berger and Prawitz, 'Inventors'.

²⁶ Key works include: Sokoloff, 'Inventive activity'; Sokoloff and Khan, 'The democratization'; Khan, *The democratization*; Lamoreaux and Sokoloff, 'Long-term change'; Lamoreaux and Sokoloff, 'Market trade'; and Lamoreaux, Sokoloff and Sutthiphisal, 'Patent alchemy'.

dissimilar to those in France, Germany, and even Britain. Our analysis is evidence that very different institutional arrangements can achieve similar outcomes.²⁷

Finally, our work sheds new light on the emergence and rapid collapse of the UKNL state. Judo and Van de Perre identify the design and functioning of the polity's legal system as a continuing lacuna in our historical knowledge of this period.²⁸ Our analysis contributes by tracking the use of day-to-day legal practices, distinct from formal 'black-letter law', particularly in the realm of innovation policy. More broadly, we help illuminate the somewhat neglected *economic* history of the Low Countries in the early nineteenth century – overlooked by Belgian and Dutch scholars alike, who tend to conclude their narratives with the Batavian Revolution of 1795, to begin with the Belgian Revolution of 1830, or to examine only the part of the UKNL that happens to fall within their country's modern borders.²⁹

We proceed as follows: Section I develops our hypotheses; Section II describes our new patent dataset and empirical strategy; Section III presents a univariate statistical analysis; Section IV reports on regression models that allow us to differentiate the data on a multivariate basis; Section V discusses results in a comparative perspective; and Section VI concludes. Several appendices complete our analysis: Appendix A details the design and administration of the UKNL patent system; Appendix B reviews the wider cliometric patent literature; Appendix C explains our occupational classification; Appendix D describes our technology classification; Appendix E presents our measure of patent quality; and Appendix F develops a conceptual model that formalises the mechanisms implied by the historical evidence in the analytic narrative tradition, which motivates our interpretation of administrative discretion.

I. HYPOTHESIS DEVELOPMENT

Drawing on the institutional and cliometric evidence presented in Appendices A and B, together with insights from the innovation economics literature, this section develops the hypotheses that guide our empirical analysis. Rather than treating patents as a general indicator of inventive activity, we focus on identifying the institutional, technological, and applicant-level factors associated with successful patent applications. Because our dataset includes both successful and

²⁷ Our argument relates to Jochen Streb's keynote lecture to the Associazione per la Storia Economica (Streb, 'Patent law'), in which he argues that nineteenth-century European patent systems were able to achieve similar outcomes to the US patent system, albeit using different institutional configurations.

²⁸ Judo and Van de Perre, *Belg en Bataaf*. Rare exceptions to this pattern include: Van IJselmuiden, *Binnenlandse zaken*; and Van den Berg, 'De integratieve functie'.

²⁹ Witte, 'Tweehonderd jaar'. Exceptions are: Horlings, 'Een batig saldo?', on the role of fiscal transfers and infrastructure investment imbalances as economic causes of Belgian independence; Buyst, 'De onmogelijke integratie', on the failed economic integration of North and South; Van de Perre, 'Union de principe', on the political economy consequences of fiscal centralisation; and Philips and Buyst, 'Regional industrialization', who begin their economic analysis in 1820 and compare both successor polities longitudinally after partition.

unsuccessful applications, it allows us to examine the determinants of patent grant decisions directly. This perspective is especially informative for patents of importation (or introduction), which were explicitly designed to facilitate the domestic introduction of technologies already in use abroad and for which administrative discretion played a central role.

Many contemporary and historical patent systems rely on self-selection through fees as a screening mechanism. By requiring applicants to incur costs at filing and during prosecution, such systems induce inventors to assess the expected future value of protection before proceeding.³⁰ The idea is that applications of low anticipated value are less likely to advance because the expected private benefits do not justify paying the required patent fees.³¹ This approach is decentralised and relatively inexpensive to administer, as it exploits applicants' superior information about their own inventions. Its limitation is that it screens on willingness and ability to pay rather than on technical merit, and may therefore exclude socially valuable inventions characterised by uncertainty or financial constraints.³²

An alternative screening mechanism is substantive patent examination, in which professional examiners assess applications against legal and scientific standards such as novelty and inventive step before rights are granted.³³ Examination shifts the burden of screening from applicants to the state, relying on specialised expertise to evaluate technical quality directly. While both fee-based self-selection and substantive examination aim to restrict patent rights to a limited set of non-trivial inventions, they differ in their informational basis and institutional cost.

The Industrial Revolution-era English patent system relied on self-selection using high fees until these were reduced in 1883.³⁴ Substantive examination was only introduced there in 1905.³⁵ The UKNL system, meanwhile, adopted a rather different approach: it charged high fees but subjected applications to substantive examination as well. Civil servants assessed technical feasibility and economic relevance, counselled applicants and sought outside advice before making their decision.³⁶ They were even empowered to offer fee discounts. Rather than granting a one-size-fits-all patent right to those applicants deemed to have met a certain patentability threshold, civil servants were authorised to use discretion over patent terms and conditions, allowing them to design bespoke patent rights for each individual applicant.

³⁰ Encaoua, Guellec and Martínez, 'Patent systems'; de Rassenfosse and van Pottelsberghe de la Potterie, 'The role of fees'.

³¹ Burhop, 'The transfer'.

³² Nicholas, 'Cheaper patents'.

³³ Hall and Helmers, *The economics*.

³⁴ Nicholas, 'Cheaper patents'; Kügler, 'The responsiveness of inventing'; Billington and Lane, 'Success and failure'.

³⁵ Billington, Colvin and Coyle, 'Financing innovation'.

³⁶ See Appendix A, in addition to Wagenaar, 'The patent system'.

This discretionary framework should be viewed in the context of the UKNL's broader approach to economic development. Differences in the economic structures and trade policy preferences of a commercial North and an industrial South led, from the foundation of the UKNL state, to difficult negotiations over economic policy that ultimately resulted in a compromise around 1822.³⁷ The government combined a relatively open trade regime with an industry fund that provided soft loans and industrial subsidies. Discretion over patent rights formed part of a wider set of policy instruments through which the state sought to encourage industrial development.

It is this discretionary feature that we examine here above all else. As discussed in Appendix A, archival correspondence surrounding the drafting and administration of the 1817 patent law suggests that this discretion was designed to enable the bureaucracy to pursue policy objectives rather than merely provide procedural flexibility. Ministers described examination as a safeguard against insignificant inventions, with correspondence emphasising that patent protection should be 'limited to the shortest possible term' necessary to reward inventors, allowing them benefits 'sufficiently as a reward' while reconciling 'general interest with individual interest'.³⁸ This principle underpinned the rejection of technically impossible or economically unpromising applications, the restriction of patentable subject matter where effective enforcement was impossible, and the adjustment of patent terms to the minimum administrators considered necessary to encourage invention and technology transfer.³⁹

A defining feature of European patent systems in this period was the close connection between patent rights and cross-border technology transfer. Governments can use patents not only to reward invention, but also as industrial policy instruments to accelerate catch-up by encouraging the introduction of foreign techniques.⁴⁰ Common mechanisms include working clauses, compulsory or open licensing provisions, and the creation of specialised patents of importation. In much of continental Europe, novelty was defined relative to the polity rather than the world, making patents of importation a distinct and widely used form of intellectual property right. The UKNL followed this tradition closely. Modelled on the French system introduced in 1791, it formally distinguished between patents of invention and patents of importation, a framework also adopted in Spain, several Italian states, and parts of Germany.⁴¹

³⁷ Brugmans, *Paardenkracht*, pp. 136–41; van Zanden and van Riel, *Nederland*, p. 135.

³⁸ 'Report to the King', NL-HaNA, 2.04.01, 4040, 17-6-1817, no. 1172 [full quote in Appendix A, translation by the authors].

³⁹ Medicines conflicted with existing pharmaceutical regulation, while food and chemical recipes were excluded because contemporaneous techniques could not reliably determine their composition after manufacture (Wagenaar, 'The patent system', p. 54).

⁴⁰ This is also true for developing countries today: Hall, 'Does patent protection help'.

⁴¹ France's 1791 law distinguished invention from importation (Nuvolari, Tortorici and Vasta, 'British-French technology transfer'), a model later followed by other countries (Armengaud, *Guide de l'inventeur*; Tolhausen, *A*

As discussed in Appendix B, Patricio Sáiz is the only other patent scholar we are aware of who has compared the use of patents of invention and importation quantitatively, finding in nineteenth-century Spain that locals and foreigners accessed both types of patents, but importation patents were shorter and more costly.⁴² He concludes that patents of importation were more likely to be taken out (1) by individuals with more production-related occupations; (2) for less complex light industry inventions; (3) for inventions that were successfully worked subsequent to being granted; and (4) by those domiciled in Spain rather than abroad.

We similarly hypothesise that patents of importation in the UKNL case differed systematically from patents of invention in both purpose and risk profile. Whereas invention patents typically covered unproven technologies, importation patents protected the commercialisation of techniques already tested abroad. Applicants were therefore entrepreneurs or intermediaries rather than inventors, and the technological uncertainty facing examiners was arguably lower. Appendix A further shows that civil servants regarded patents of importation as serving a distinct economic purpose. Rather than granting exclusive rights automatically, they considered whether monopoly protection was really necessary to encourage the domestic introduction of foreign technologies and adjusted patent terms and conditions accordingly.

This institutional logic suggests that, all else equal, patents of importation should have enjoyed a higher probability of acceptance than patents of invention as they were more likely to have been considered proven technologies that were commercially viable enough to warrant importing into a new jurisdiction. At the same time, their policy rationale made them more directly subject to discretionary assessment and potential systematic discrimination. Appendix A discusses how authorities frequently shortened the lifespan of patents of importation below the legal maximum determined by the remaining term of the patent protecting the same invention abroad because they judged the imported technologies to be less effortful and so necessitated fewer incentives. And while importation patents were subject to the same working requirements as invention patents, correspondence suggests that these were enforced more stringently. Officials also retained the right to impose compulsory or open licences.⁴³

The discretionary nature of the UKNL's patent system suggests that the likelihood of success could depend not only on formal eligibility, but also on assessments of economic usefulness, effort, and expected contribution to domestic development. These assessments were

synopsis) including Spain (Pretel, *Institutionalising patents*; Sáiz, 'Did patents'), some Italian states (Nuvolari and Vasta, 'Independent invention'), parts of Germany (Donges and Selgert, 'Technology transfer'), and, from 1817, the UKNL (Wagenaar, 'The patent system'). See further discussion in Appendix B.

⁴² Sáiz, 'The Spanish'; Sáiz, 'Did patents'.

⁴³ Open licence clauses, rejection of 'easy' importations, working clauses, and reductions in patent term that developed in the UKNL system remained common practices in post-independence Belgium (Varlet, *Recueil des lois*, pp. 28–32).

inherently context-specific and raise the question of how technological and industrial conditions shaped approval decisions. Patents of importation could have served different policy objectives depending on the sector concerned. In industries with little prior domestic production, granting a patent of importation may have supported infant industries by providing temporary protection during an initial learning phase. In more established but technologically lagging industries, such patents may instead have been used to accelerate modernisation by facilitating the adoption of superior foreign techniques. Market structure may also have been relevant, since granting exclusive rights in already concentrated industries risked reinforcing existing market power. These alternative logics imply heterogeneous treatment across technologies and industries, but do not generate an unambiguous prediction regarding its direction. This is therefore an empirical question.

These discretionary judgements were mediated through the examination process, in which outside advisers with industrial or technical expertise played an important role. The fact that the advice of industrialists could be sought as part of the UKNL's formal patent application process meant that incumbent producers may have had a means to influence an application's outcome. The UKNL's substantive examination did not formally guarantee novelty or utility, but the language of correspondence between civil servants and applicants suggests that these were exactly the features being assessed.⁴⁴ Archival correspondence suggests that advisers evaluated applications in light of both technical merit and their likely contribution to domestic industry.⁴⁵ Taken together, these practices suggest that patents of importation may have served as an instrument of technology policy, with exclusive rights calibrated to the perceived need to encourage the domestic introduction of foreign technologies. Whether this policy complemented or substituted for domestic inventive activity remains an open empirical question.

Issues of accessibility and discrimination further shape the success of applications. The literature has debated whether early nineteenth-century European patent systems were socially elitist or broadly accessible.⁴⁶ But recent work suggests that they relied primarily on self-

⁴⁴ The UKNL sought to avoid patents being interpreted as state certification of an invention's quality. It adopted wording from French patent legislation, which lacked a preliminary examination (see, e.g., 'The report to the King on resistance by Glaser to the patent request by Heinisch', NL-HaNA, 2.04.01, 4451, 2-3-1829, 203A). For wider discussion about patents and signalling, see Billington, Colvin and Coyle, 'Financing innovation'.

⁴⁵ This policy of making the patent lifespan reflect the effort of the applicant and the likelihood that they would make a decent but not excessive profit – and therefore to constrain patent term more often for importers – was formulated in several reports to the King in June 1817 (e.g., 'Report to the King on patent request Berthelin', in NL-HaNA, 2.04.01, 4039, Dossier Berthelin, 23-6-1817, 1218).

⁴⁶ See especially the debate between Khan, *The democratization*, and Streb, 'Inventing ideas', on whether European patent systems were as democratic as that of the US. Berger and Prawitz, 'Inventors', are dissenters on the European side: Sweden's patent system was not accessible to non-elites (see Appendix B).

selection by charging high fees and informal screening rather than on overt social exclusion.⁴⁷ We expect the UKNL to resemble other continental systems in being broadly accessible, while still allowing for differential outcomes across applicant groups. The UKNL's system could have been democratic, despite its high fees, as reduced rates were available to those who claimed insufficient means; essentially, application fees were means-tested.⁴⁸

Internal discrimination is also plausible. The recent political unification of the Northern and Southern Netherlands into a single jurisdiction, combined with a highly centralised bureaucracy dominated by Northern elites, raises the possibility that applicants from different regions or linguistic backgrounds faced unequal chances of success.⁴⁹ The UKNL state has been portrayed as exploiting the South as a revenue-generating territory to service sovereign debt held by Northern elites.⁵⁰ Because officials set patent fees and terms on a case-by-case basis, application outcomes provide multiple observable margins on which such differences could appear. In addition to outright approval or rejection, differential processing time – the interval between application and decision – constituted a further, informal channel of discrimination, as delaying patent grants could weaken applicants' commercial prospects.⁵¹

In what follows, we analyse the determinants of successful patent applications using grant decisions, application fee discounts and processing times, and the divergence between requested and granted patent terms. Particular emphasis is placed on patents of importation. By examining how success probabilities vary across technologies, industries, market structures, applicant origins, and regions, and by comparing importation patents with invention patents within the same technological fields, we can assess how administrative discretion and industrial policy objectives shaped the practical likelihood of securing patent protection in the UKNL. This approach moves beyond abstract debates about patent 'democracy' and instead evaluates how patent institutions operated in practice as selective instruments of technology transfer in the early nineteenth-century Low Countries.

⁴⁷ Bottomley, 'Patents'; Baudry, 'Examining inventions'; Galvez-Behar, 'The patent system'.

⁴⁸ Before any discounts were applied, the full statutory cost of a five-, ten- or fifteen-year patent was 150, 300 or 600 Dutch guilders. By comparison, the daily wage for an unskilled worker in the Northern Netherlands in 1819/1820 was one guilder (van Zanden and van Riel, *Nederland*, p. 84).

⁴⁹ Van IJsselmuiden, *Binnenlandse zaken*; van Zanten, *Schielijk*; Deneckere, 'Continuity and discontinuity'. The overall share of Southern Netherlands in government was just 10%. However, 40% of civil servants hailed from the South in the department responsible for the patent system. 3 million people lived in the Southern Netherlands, and 2 million in the Northern Netherlands (Commissie voor de Statistiek, *Opgave der bevolking*).

⁵⁰ Buyst, 'De onmogelijke integratie'; van Zanden and van Riel, *The strictures*.

⁵¹ Such administrative delays have been documented as a discriminatory device in several German states: Lehmann-Hasemeyer and Streb, 'Discrimination'.

II. DATA AND EMPIRICAL STRATEGY

We constructed a dataset of all patent applications in the UKNL between 1817 and 1830 in four stages. First, we extracted a dataset from Doorman's list of all granted patents in the Netherlands in the nineteenth century.⁵² Doorman's data were completed and corrected using the registry of all patents granted between 1817 and 1869, obtained from the Nationaal Archief in The Hague.⁵³

Second, we used the ledgers of all communications and dossiers of the Ministry of Education, the government department responsible for the system's administration, to reconstruct all applications related to inventions.⁵⁴ This allowed us to retrace the full sequence of procedural steps that each patent application underwent – for example, whether and how often it was returned to applicants for correction, from which we can infer the length of the application procedure for each patent application. The successful patents were cross-referenced with the dataset of granted patents. Crucially, using these records also enabled us to retrieve applications that were discontinued or rejected, as well as those of inventors who sought loans or prizes instead of patents.

In the third stage, all dossiers retrievable in the relevant archives were searched for important details such as the title or nature of the invention, the terms for which the applicant had applied, the applicant's occupation, and the reasons for rejection.⁵⁵ In the fourth stage, we categorised patentees and patents to aid our analysis. The occupations were encoded into HISCLASS categories following the procedure outlined by van Leeuwen and Maas, as used recently by Berger and Prawitz for Swedish patentees.⁵⁶ The classes were then combined into larger groups to avoid relatively arbitrary class distinctions due to the exact choice of terms used by applicants to describe themselves. The patent titles were categorised into technological classes based on the 20 categories developed by Billington and Hanna.⁵⁷ For maximum accuracy, we first manually translated any applications in French into Dutch to have a uniform language, then adapted their machine-learning tool to Dutch to classify our patents, and finally

⁵² Doorman, *Het Nederlandsche octrooiwezen*.

⁵³ The UKNL's patent registries, including Doorman's replacement registry, are held by the Nationaal Archief (Dutch National Archives) in The Hague under NL-HaNA, 2.04.23.01 (Ministerie van Binnenlandse Zaken: Afdeling Nijverheid en voorgangers, 1817-1877), 1236-1240 (Registers van verleende octrooien).

⁵⁴ For ledgers, see NL-HaNA, 2.04.01 (Ministerie van Binnenlandse Zaken, 1813-1848), 4049-4051 (Indices 1815–1818); 4190-4196 (Indices op de verbalen 1818–1824); 4925-4936 (Indices op de verbalen 1824–1830). Photographs are available digitally: <https://www.nationaalarchief.nl/en/research/archive/2.04.01>.

⁵⁵ See NL-HaNA, 2.04.01, 4039-4942 (Uitvindingen); 4055-4175 (verbalen: besluiten, minuten van uitgaande brieven en ingekomen stukken, 1818–1824); 4204-4545 (verbalen 1824–1830).

⁵⁶ Van Leeuwen and Maas, *HISCLASS*; Berger and Prawitz, 'Inventors'. See Appendix C for our HISCLASS encoding approach, descriptive statistics of the outcome, as well as a comparison with the HISCLASS coding for patentees in England.

⁵⁷ Billington and Hanna, 'That's classified!'. See Appendix D for further discussion, including summary statistics by technology class.

manually corrected the output. Additionally, as an alternative classification designed to capture the core technology of the Industrial Revolution, we manually identified all patents for steam-related inventions, defined as patents whose title claimed either to improve steam engines or to use steam engines for another purpose (such as textile machinery or transportation).

Finally, we used Doorman's technology histories to build a patent quality indicator for granted patents. Doorman, who had worked for the Dutch patent office as an examiner during the first half of the twentieth century, wrote his technological history of the Netherlands through the lens of the country's patent system as his retirement project. Using a wide range of bibliographical material from the nineteenth century as well as registries found or reconstructed by him in the archives, he accompanied his list of granted patents with individualised commentary.⁵⁸ He frequently inserted references to individual patents or patentees found in the contemporary literature or the technological historiography of his time, or to prizes won at industrial fairs. We used these comments to build a 'Doorman indicator' for valuable patents.⁵⁹

Our resulting dataset is very rich. It contains details of all patent applicants, including their name, place and province of residence within the UKNL, country of residence if foreign, and occupation. It also includes information about the invention being applied for, including an *ex post* quality indicator, patent type, patent lifespans and fees requested, the lifespan, fees and conditions imposed on the applicant, each administrative step linked to the relevant date, which external bodies were involved in the processing and examination of the application, whether the patent application had been rejected and by whom, and whether the patent had been transferred to another owner subsequent to grant and to whom. The variables created from this database for this article are defined in Table 1.

[INSERT TABLE 1 HERE]

[INSERT FIGURE 1 HERE]

[INSERT FIGURE 2 HERE]

Figure 1 reports the patent application success ratio across time and shows the growth in the system's use across our period of study. Figure 2 is a flowchart depicting how the Dutch patent system functioned in this period.⁶⁰ It shows how the patent system fitted into the UKNL's wider national innovation system; the archival files also record requests for subsidies, income

⁵⁸ Doorman, *Octrooien voor uitvindingen*; Doorman, *Het Nederlandsche octrooiwezen*; Doorman, *Techniek en octrooiwezen*.

⁵⁹ The indicator is a binary (dummy) variable constructed after excluding citations that Doorman regarded as self-promotion by the patentee and patents he discussed solely out of personal interest; Appendix E provides further details.

⁶⁰ Appendix A uses this figure to describe the institutional design of the UKNL patent system; Appendix F formalises the process in a conceptual model.

support and loans, as well as requests for honorary titles and tax benefits. Requests often overlapped; several individuals requested both a premium and a patent, or asked for a subsidy but were instead forwarded to the patent authorities. In this article, we focus only on the 1,093 patent requests, thereby excluding other policy interventions from our core analysis.

The second important distinction evident from Figure 2 is whether a patent request is for a patent of invention, improvement or importation. Elsewhere, Wagenaar explains the legal and institutional features of these different types of patent rights.⁶¹ In short, ‘invention’ covered patents for new inventions by domestic inhabitants; ‘improvement’ signified improvements added to existing granted patents that were not deemed sufficiently novel to warrant a stand-alone patent; and ‘importation’ applied to the first person to import or introduce a technology into UKNL territory, whoever the original inventor was.⁶²

[INSERT FIGURE 3 HERE]

[INSERT FIGURE 4 HERE]

Figure 3 maps political borders, population and industrial employment in 1819 across three panels for reference. Figure 4 maps our dataset of patents of invention in two panels. A comparison of these maps, which depict patent applications and grants, does not suggest systematic discrimination across regions by patent officials. A visual inspection suggests that grants appear to number roughly in proportion to applications. Overall, the pattern of patenting corresponds to the pattern of industrial employment, but with some important outliers. The strongest industrial employment in 1819 is found in West-Vlaanderen (WVLA) and Oost-Vlaanderen (OVLA), yet patenting rates there across 1817–1830 are low. These provinces had a widespread rural textile industry (linen) that employed a large share of the population, using traditional production methods. Only Ghent, the capital of Oost-Vlaanderen, had a flourishing Manchester-style cotton industry, thanks to successful early industrial espionage.⁶³ The other outliers are the strong patenting performance of Zuid-Holland (ZHOL) and Zuid-Brabant (ZBRA), neither of which were industrial centres at the time of the 1819 census – though both contained the UKNL’s alternating capital cities.⁶⁴

⁶¹ Wagenaar, ‘The patent system’.

⁶² Due to their small number and the fact that they were always filed by the original patent holder, we exclude improvement patents from our main analysis. We also count as withdrawn those applications that never progressed far enough to be formally classified by type.

⁶³ Buyst, ‘The causes’.

⁶⁴ Strong patenting rates in capitals have also been found in other studies (e.g., Nuvolari, Tortorici and Vasta, ‘British-French technology’). They are more likely to reflect patent applicants’ and agents’ preference for proximity to the administrative process than any inherent patenting propensity of the local population.

To explore the drivers of patent success, we employ an empirical strategy that examines whether statistical associations exist between various indicators of interest within the context of the UKNL patent system. First, we employ univariate statistics that juxtapose the means of the various variables of interest directly across divisions of patent type, including granted and ungranted patents, and geography. Then, we compute multivariate statistics to take account of possible interacting effects between variables. We do this by using discrete choice (logistic) models to estimate what indicators seem best to ‘predict’ patent type, patent grant decisions, the conditions under which the patents were granted, and the value of patents.

In addition to reporting a pseudo- R^2 statistic to assess overall model fit, we evaluate predictive performance using the area under the receiver operating characteristic curve (AUC). The receiver operating characteristic (ROC) curve plots the true positive rate (sensitivity) against the false positive rate ($1 - \text{specificity}$) across all possible classification thresholds, and the AUC provides a summary measure of a model’s ability to discriminate between positive and negative outcomes. An AUC of 0.5 indicates no discriminatory power beyond random classification, while values closer to 1 imply increasingly accurate discrimination.⁶⁵ Originally developed in the biomedical sciences, the AUC has become a standard diagnostic tool for evaluating the predictive performance of logistic regression models across a wide range of applied research settings.⁶⁶

III. UNIVARIATE STATISTICAL ANALYSIS

Table 2 reports the univariate statistics for the full sample. Panel A shows that of all patent applications, 48 per cent were eventually granted. About 22 per cent of applications were withdrawn or discontinued after the state informed applicants that their request did not meet the legal requirements. Most applications were submitted by a single applicant, with only 11 per cent submitted by multiple applicants. This suggests that for most inventors or importers, a partnership with a wealthier individual was not required to gain access to the patent system.⁶⁷ The average time to decision was about 3.7 months, although with considerable variation across applications.

[INSERT TABLE 2 HERE]

⁶⁵ A rule of thumb for interpreting these statistics is due to Hosmer, Lemeshow and Sturdivant, *Applied logistic regression*: a value between 0.5 and 0.7 means that the model performs poorly; a value between 0.7 and 0.8 is acceptable; one between 0.8 and 0.9 is excellent; and one between 0.9 and 1 is outstanding.

⁶⁶ For example, it is used by Colvin, de Jong and Fliers, ‘Predicting the past’, to analyse the causes of bank failures.

⁶⁷ Dutch law required a sponsor either to be listed as an inventor or to register following a patent transfer (Wagenaar, ‘The patent system’). Only 17 patents were transferred in our study window.

The data show that only a small proportion of patents (19 per cent) were requested by someone who had applied before during the same period. By contrast, a much larger share (42 per cent) of American patents from a similar period (1812–1830) were requested by patentees who held more than one patent.⁶⁸ If we follow Sokoloff and Khan’s logic, then the relatively low number of career patentees suggests a democratic use of the patent system. Around 30 per cent of patent applications were examined by a technical adviser, while only about 4 per cent were forwarded to Chambers of Commerce and Industry for an opinion on novelty. Only a small share of applications (7 per cent) concerned steam-related technologies, reflecting the fact that such inventions constituted an important but still limited subset of patenting activity.

Patent requests for inventions are distinguished from importations in the right-hand columns of Table 2 Panel A. The intuition that these patent types are treated differently is supported: a test of means detects a statistically significant difference in almost all variables. Consequently, we treat both patent types separately in regression analyses. The two indicators that are similar are the share of applications withdrawn after their first confrontation with the legal standards, and the proportion of applicants with several patent applications.

Interestingly, patents of importation are more likely to be refused than patents of invention, which is consistent with the UKNL’s policy of tougher investigations into this type of patent, as described by Wagenaar.⁶⁹ Besides having to conform to the usual standards of novelty and utility, the practice was to grant patents of importation only for inventions that would not transfer easily across countries without patent rights. The data confirm that industrial advisers, who were asked to gauge market conditions, were consulted more frequently in the examination of patents of importation. For patents of invention, the bureaucracy was more likely to involve technical advisers, in line with the expectation that novel inventions require more technical competence to judge their novelty and utility. Patents of importation are also somewhat more likely to involve multiple applicants and steam-related technologies.

Applicants for patents of importation tend to be better connected. They are more likely to work in teams and more likely to come from higher socioeconomic classes – whether this means a merchant-related position, a highly trained professional, a manufacturer, a military officer or a nobleman. Meanwhile, applicants for patents of invention are most likely to come from the artisanal classes, whose inventions emerge directly from their own occupations – what Mokyr calls the ‘tinkerers’.⁷⁰ In particular, medium- and lower-skilled workers are significantly better

⁶⁸ Sokoloff and Khan, ‘The democratization’. Similar figures held for France in these years (Nuvolari, Tortorici and Vasta, ‘British-French technology’).

⁶⁹ Wagenaar, ‘The patent system’.

⁷⁰ Mokyr, *The enlightened economy*.

represented among invention patents, while factory owners, nobles and landholders are more strongly represented among patents of importation.

In Table 2 Panel B, we split the data according to whether a patent was granted or not. Applications that eventually received a patent tended to take longer to process than those that were rejected, suggesting that successful applications were subject to more extensive evaluation. For both patents of importation and invention, a shorter distance to one of the two administrative capitals of the UKNL appears to increase the likelihood of a patent grant.⁷¹ An industrial adviser was consulted with equal probability for granted and ungranted patents, whether importation or invention. In contrast, a technical adviser was consulted more frequently in cases where applications were granted, for patents of either type. This implies that the state was willing to refuse patents without technical advice. Manufacturers, nobility and landholders were more likely to be granted patents of invention or importation. For patents of importation, the merchant class is also well represented. This is consistent with the hypothesis that better-connected individuals are more likely to import foreign technology. Applicants with prior experience in the patent system are more likely to obtain patents of importation. The opposite holds for higher professionals.⁷²

Table 2 Panel C summarises the data by the geographic origin of applicants. On the left side, we split the data according to the UKNL's internal geography. We see that southern applicants were (1) relatively more likely to apply for patents of importation and less likely to apply for patents of invention; (2) somewhat more likely to be successful patentees; and (3) more likely to come from the manufacturing, landholder and nobility classes. By contrast, northern applicants are more likely to hail from the merchant class. This contrast between manufacturers and commerce is consistent with the idea of a manufacturing South and a commercial North.⁷³ Applicants in the North also appear, on average, to live further from the administrative capitals, reflecting the geography of the kingdom. No statistical difference is found in the frequency of involvement by technical or industrial advisers between the two parts of the kingdom, suggesting that the patent system was administered uniformly across the UKNL.

We compare domestic requests for patents of importation with foreign requests for patents of importation on the right side of Table 2 Panel C. These univariate comparisons suggest there

⁷¹ The capital alternated between Brussels and The Hague every October. Over 800 civil servants, politicians and their families moved at this time (van IJsselmuiden, *Binnenlandse zaken*).

⁷² Applicants whose class we cannot discern are more likely to have had their patent rejected. We think this is a mechanical result because successful applicants engage with the patent process for longer, and thus more records about them remain in the archives, making it more likely that their profession can be ascertained.

⁷³ Mokyr, *Industrialization*.

was likely no discrimination in patent grant decisions. Foreign applicants are significantly more likely to be highly trained professionals, while domestic applicants are somewhat more likely to come from artisanal or manufacturing backgrounds. This may be a consequence of a more advanced state of professionalisation in France and the UK, from which most foreign applicants hailed, than in the UKNL.

[INSERT TABLE 3 HERE]

Table 3 reports the univariate statistics for the full sample of granted patents, as well as for patents of invention and importation separately, again with tests of differences in means between the two patent types. We list various dimensions along which the UKNL state could discriminate: (1) the time taken to process patent applications; (2) the difference between the term of patent protection requested and assigned in years; (3) the size of any patent application fees levied; (4) whether additional conditions were assigned to granted patents, such as open licence clauses or a reduced length of the working clause.

The statistics suggest that the state discriminated between the two patent types on most fronts. The only exception appears to be the assigned patent term. However, the assigned lifespan differs significantly from the requested term. Patents of importation requested slightly longer terms on average than patents of invention and were more likely to have shorter terms imposed on them by the patent bureaucracy. About 44 per cent of patents of importation had their lifespan reduced following examination, compared with 25 per cent of patents of invention. Fee reductions show the opposite pattern: about one third of patents of invention received a fee reduction, compared with just 12 per cent of patents of importation. Patents of importation were also more likely to be granted with special conditions attached and took slightly longer to process on average.

For the successful patents, we observe a familiar pattern: the highest class of manufacturers and nobility are more frequently patentees for imports, while medium-skilled and lower-skilled workers and artisans are more frequently granted patents for inventions. The higher professionals and merchant classes are relatively evenly split between patents of invention and importation. Patents of importation also tend to originate from applicants located further from the UKNL's administrative capitals and are somewhat more likely to concern steam-related technologies.

The average requested term of almost 11 years is high, considering that a ten-year patent would have cost the entire annual income of an unskilled worker. It likely reflects the wealth of the average patentee, or the expectation that they would be able to obtain a patent for a reduced fee, or even for free, by claiming insufficient financial means. Our Doorman indicator

suggests patents of importation were relatively more likely to have a higher *ex post* value than patents of invention: 16 per cent of patents of importation have a higher value, compared with 10 per cent of patents of invention – an outcome that conforms to the expectation that valuable patents are patented across more jurisdictions and that patents of importation served as an important means of facilitating technology transfer.

IV. MULTIVARIATE STATISTICAL ANALYSIS

While useful for descriptive purposes, univariate statistics do not capture the interdependencies between variables. To address this, we employ regression analyses to explore statistical regularities within a multivariate framework. We begin by examining the differences between patents of importation and patents of invention. We then investigate which types of patent applications were more likely to result in granted patents. We finish by looking at patent grant conditions and patent value.

Correlates of Invention vs Importation

Table 4 reports a set of logistic regressions examining whether there is a systematic distinction to be made between patents of invention and patents of importation. The first four regressions are for the full sample of all applications, while the last four use the sample of granted patents.

[INSERT TABLE 4 HERE]

Each group of four regressions follows a regular pattern in that they include the country of origin and the social class as control variables. Across all regressions we control for time period of application and technological class of the invention. We include a time dimension to account for any changes to patenting rates and practices across our period. We include technology classes as controls because we know from other work that technology fields have different propensities to patent and use the patent system differently.⁷⁴ We also include a specification which replaces technology fixed effects with a dummy for steam-related technologies. The R^2 and AUC statistics indicate that the models' predictive quality improves significantly when geographical and class dimensions are included.

First, the table confirms that the patent bureaucracy treats applicants for patents of importation and patents of invention quite differently. Patents of importation are significantly less likely to be granted in the full sample, indicating that such applications faced a lower probability of success. Among granted patents, patents of importation are associated with

⁷⁴ See, e.g., Moser, 'How do patent laws'.

longer requested terms and are significantly less likely to receive fee reductions, suggesting that the financial and temporal conditions attached to these patents differed systematically from those granted for inventions. While technical advisers are involved in adjudicating both types of patents, industrial advisers are significantly more likely to be associated with patents of importation, pointing to the greater commercial and industrial relevance of imported technologies. Applications involving multiple applicants are also more likely to concern patents of importation, consistent with the idea that these patents were often linked to organised commercial ventures. Prior experience with the patent system does not appear to systematically distinguish between the two types of patents.

Additional differences emerge in the characteristics of the patents themselves. Steam-related technologies are significantly more likely to be associated with patents of importation, reflecting the role of technology transfer in sectors central to the Industrial Revolution. Among granted patents, those associated with the Doorman indicator – our proxy for *ex post* technological significance – are also more likely to be patents of importation. Time to decision is weakly positively associated with patents of importation in some specifications, suggesting that these applications either required longer administrative scrutiny or were discriminated against in the administrative process.

In terms of geography, the table confirms that individuals residing in the South are more likely to obtain patents of importation than those in the North, even when the social classes variable is included, which likely reflects market opportunities gained during industrialisation. Foreign patentees were much more likely to obtain patents of importation since they were legally required to do so; only a small minority of such applicants were foreigners settled in the UKNL or partnered with domestic parties.

The class of factory owners, nobility and landholders is our reference category. Other higher and wealthier classes behave quite like them in their demand for patents of importation. It is the artisanal class, as well as medium- and lower-skilled workers, who are less likely to ask to import technology and more likely to invent it themselves.

Overall, our multivariate analysis is consistent with the idea that patents of invention are distinct from patents of importation in terms of their characteristics, the characteristics of their applicants, and their treatment by the state. For this reason, in the remainder of this article we investigate each category of patents separately rather than including both application types in the same regression framework.

Correlates of Successful Patent Grants

Table 5 further demonstrates what factors are systematically associated with patents granted by the UKNL's patent authorities. The first three models show the success rate for patents of invention, while the last three focus on patents of importation. We include only those patents that have 'passed' the initial legal test and proceeded with a full application. Withdrawn patents are excluded so that we can more clearly observe the patent bureaucracy's discretionary decision to grant or reject applications that reached the substantive evaluation stage.

[INSERT TABLE 5 HERE]

Interestingly, the table suggests little evidence either of systematic discrimination in patent grant decisions for patents of invention or that the patent system was inaccessible to applicants once they could supply sufficiently accurate patent specifications.⁷⁵ In predicting the success of a patent application, most applicant characteristics – such as geography, social class, or prior experience with the patent system – do not appear to matter. Almost nothing is statistically significant. The main exception is distance to the capital, which is negatively associated with the probability of a patent being granted. The R^2 and AUC indicate that the models' fit or accuracy improves only marginally when geographical or class indicators are included.

The story is different for patents of importation. Here several variables are significantly associated with success. Distance to the capital is strongly and positively associated with the probability that a patent is granted, and applications involving multiple applicants are more likely to succeed in some specifications. By contrast, the involvement of a technical adviser is negatively associated with success, suggesting that technically scrutinised applications were more likely to be rejected. Higher professionals and applicants with unknown class are also significantly less likely to obtain patents of importation than the reference group of factory owners, nobility, and landholders. Nevertheless, there is no clear evidence of discrimination along either internal or external borders, as the coefficients for southern and foreign applicants are not statistically significant.

Correlates of Granted Patent Conditions

We test for the correlates of patent fee reductions, reductions in the assigned patent term, and the time taken to reach a decision in Table 6. This table reports logistic regressions for patent

⁷⁵ Those applicants who claimed they did not have the capacity to make accurate technical drawings and descriptions were assisted by authorities. See, e.g., 'Invitation to the governor of North-Holland to invite the burgomaster of Amsterdam to appoint an expert who can help P. Elders to improve the drawings of his invented hydraulic engine' ('Information to Elders', NL-HaNA, 2.04.01, 4105, 28-12-1820, 1928).

fee reductions in models 1 and 2, distinguished by patent type, and for reductions in the patent term assigned at the point of grant in models 3 and 4. Models 5 and 6 report OLS regressions for the time taken, in months, to reach a decision on patent applications.

[INSERT TABLE 6 HERE]

The results for patent fee reductions suggest that these operated as a form of implicit means testing, with fee discounts more closely related to the financial position or characteristics of applicants than to the technological qualities of the invention itself. The social-class indicators reinforce this interpretation: medium- and lower-skilled applicants are more likely to receive fee reductions for patents of invention than factory owners, nobility, and landholders. There is little association between the *ex post* value of an invention and the reduction of patent fees for patents of invention, although the Doorman indicator is positively associated with fee reductions for patents of importation.

The relationship with distance to the closest capital is negative and statistically significant for patents of invention, indicating that applicants living further away from one of the UKNL's capitals were less likely to obtain a reduced patent fee.⁷⁶ One possible explanation is informational rather than discriminatory. Applicants residing closer to the administrative centres may have been better informed about the possibility of requesting reduced or waived patent fees, either through personal contact with officials or through local networks.

Applications involving multiple applicants and those submitted by experienced patentees are also less likely to receive fee reductions for patents of invention, which is consistent with teams of applicants including investors capable of financing the patent. For patents of importation, the involvement of an industrial adviser in the assessment of an application is positively associated with fee reductions, while the involvement of a technical adviser is negatively associated with them.

The decision to reduce the patent term, reported by models 3 and 4, has a much less clear set of correlations. Few variables are statistically significant. Patents examined by a technical adviser are somewhat less likely to receive a shorter patent term for patents of invention, while applicants of unknown social class are somewhat more likely to receive shorter patent terms. For patents of importation, neither the involvement of industrial advisers nor that of technical advisers appears to matter.

⁷⁶ The possibility of obtaining a free patent was not mentioned in the law; the existence of this practice appears to have spread by word of mouth. In 1828, civil servants at the ministry complained to the governor of South Brabant that applicants from Brussels were requesting free patents as a matter of course, suggesting that knowledge of this practice was particularly widespread in the co-capital (see 'Letter to the Governor of South-Brabant on the request of Meulemans', NL-HaNA, 2.04.01, 4439, 18-12-1828, 41F).

Finally, the OLS regressions for time to decision, reported in models 5 and 6, suggest that the duration of the administrative process was only weakly associated with observable characteristics of applicants or patents. Applications requesting longer patent terms take significantly longer to process for patents of invention, suggesting that administrators devoted more scrutiny to applications seeking longer protection. Some applications involving multiple applicants take longer to process for patents of invention, while experienced applicants tend to receive decisions more quickly for patents of importation. The involvement of a technical adviser is also associated with longer processing times for patents of importation, suggesting that technically complex cases required more extensive administrative scrutiny. We believe there is no clear indication of discrimination in the timing of administrative decisions.

Correlates of Patent Value

In our last set of regressions, reported in Table 7, we explore the correlates of the Doorman indicator, our *ex post* indicator of patent value derived from technology histories. Because the number of higher-value patents is rather limited, we do not find many significant associations. However, there does appear to be a link between applicant skill and patent value, in the sense that higher professionals are significantly more likely to have obtained a high-value patent than the class of factory owners, nobility and landholders. This is consistent with the expectation that skilled patentees submitted more economically valuable inventions for consideration.

[INSERT TABLE 7 HERE]

By contrast, applicants drawn from the merchant and clerical classes appear less likely to obtain high-value patents of importation. Applications involving multiple applicants or those submitted by experienced patentees are also significantly less likely to correspond to high-value patents of importation, suggesting that commercial organisation or familiarity with the patent system did not necessarily translate into technologically significant inventions. There appears to be no strong statistical association between patent value and the place of residence of the applicants.⁷⁷

V. DISCUSSION

We have analysed, for the first time, the functioning of the patent system of the UKNL, a short-lived polity that uniquely encompassed both an early-industrialising region (Belgium) and a

⁷⁷ See Appendix E for a discussion of patent value by technology class and region of applicant domicile.

late-industrialising one (the Netherlands). This institutional setting allows us to examine how a single patent framework operated across markedly different economic environments.

We find that the system's in-built distinction between patents of invention and patents of importation created two fundamentally different instruments, which differed not only in their legal and administrative treatment by government officials, but also in the characteristics and motivations of their applicants. Patents of importation were sought by applicants aiming to transfer existing technologies that had already proven viable abroad and therefore involved less technological uncertainty, whereas patents of invention were associated with higher levels of technical skill, investment, and engagement with domestic production. These differences were explicitly recognised by administrators, who exercised discretion to differentiate between the two categories when setting fees, terms, and conditions, aligning patent policy with their underlying economic functions rather than applying uniform rules.

Indeed, our cliometric evidence suggests that the UKNL state's patent bureaucracy exercised discretionary powers to balance intellectual property rights granted to individual technology importers with the interests of wider society, ensuring that patents hindered the economy as little as possible. Such discretion, which appears to have been intended to ensure equal treatment, proportionality and fair play, is consistent with Dutch legal culture as described by Van Rossum.⁷⁸ Our case suggests that discretion *can* be time-consistent when it is exercised systematically to pursue a clear policy objective.⁷⁹ Importantly, we also find no evidence that the administrative processing time of patent applications was used as a tool of discrimination, in contrast to the case of Württemberg.⁸⁰

Instead, the regression results indicate that administrators calibrated patent terms in response to the characteristics of the application itself. Applications requesting longer patent terms were less likely to receive fee discounts and took longer to process, implying that the requested lifespan was a key consideration in determining patent terms. Administrators also appear to have recognised that inventing new technology required greater effort than importing existing technology and sought to identify under what conditions a patent of importation was necessary to stimulate technology transfer. Applicants for patents of importation were typically more commercially minded and well connected, while applicants for patents of invention

⁷⁸ Van Rossum, 'Dutch legal culture'.

⁷⁹ This contrasts with the theoretical argument of Kydland and Prescott, 'Rules rather than discretion', that discretion can give rise to time inconsistency when policymakers have incentives to depart from previously announced policies, making credible rules preferable even when they represent only a second-best solution.

⁸⁰ Lehmann-Hasemeyer and Streb, 'Discrimination'.

tended to possess stronger technical skills and competencies. This finding is consistent with those for Spain, but contrasts with those for France.⁸¹

The characteristics of patents of importation suggest that this institution played a meaningful role in the diffusion of new technologies. Twice as many were sought by domestic citizens as by foreigners, indicating that local entrepreneurs actively used the system to introduce foreign technologies into the UKNL economy. The industrialising Southern Netherlands also made greater use of patents of importation, suggesting that this instrument was particularly important there. That patents of importation plausibly had a higher value than patents of invention reinforces this interpretation. The actual technology transfer from Britain and France was likely greater than we can observe from the patent system's use.⁸² Our patent data include applications from several English and French nationals living in the UKNL's South, illustrating how the patent system complemented other channels of international technology transfer.⁸³

We find no strong evidence of geographical discrimination. The UKNL patent system appears to have processed requests from the Northern Netherlands, the Southern Netherlands and abroad equitably. This finding holds whether we use the indicator of patent grant rates, the conditions under which the patents were granted, or application processing speeds. Of course, patents of importation were treated differently from patents of invention; the bar for proving both the utility and the novelty of such inventions was set higher, and the conditions imposed were stricter. However, this applied to domestic as well as foreign applicants. Considering that foreign applicants would normally already have been able to obtain and commercialise an equivalent patent in their own country, their position is different from that of a domestic inventor. The only possible discrimination we detect is that discounted patent fees were rarely granted to foreigners, while even a domestic importer could sometimes obtain a free patent. Aside from this, the broadly fair treatment of applicants places the UKNL close to France and Baden, and distinguishes it from Württemberg.⁸⁴

The social background of patentees provides insight into how the UKNL patent system functioned. The system was accessible to non-elite applicants, with a large share of applications

⁸¹ Sáiz, 'Did patents'; Nuvolari, Tortorici and Vasta, 'British-French technology'.

⁸² The Southern Netherlands constituted an attractive labour market for skilled workers from Britain, who brought with them technical knowledge and industrial expertise, illustrating that patenting complemented broader channels of international technology transfer (Lefebvre, De Caigny and Buyst, 'English industrial workers').

⁸³ In 1835, a national industry exhibition evaluated who had contributed the most to Belgian industry and awarded two machine-builders: Huyttens-Kerremans, who owned patents of importation, and Cockerill, who instead imported and copied machines without using patents. It also rewarded de Bast de Hert for importing and popularising English cotton spinning technology, also without using patents (Gachar, *Rapport du juri*).

⁸⁴ France: Nuvolari, Tortorici and Vasta, 'British-French technology'; Baden: Donges and Selgert, 'Technology transfer'; Württemberg: Lehmann-Hasemeyer and Streb, 'Discrimination'.

and grants coming from artisans, small business owners, and medium-skilled workers, who were also the most likely to receive discounted or free patents. Participation from the lowest social classes remained limited, however, likely reflecting lower awareness of the patent system and of the availability of fee reductions, although their participation was broadly comparable to that observed in England during the same period.⁸⁵

Skilled individuals – particularly engineers and other technically trained professionals – nonetheless accounted for a disproportionate share of patents that later proved economically valuable, consistent with the findings of Alessandro Nuvolari and co-authors that highly skilled inventors were more likely to produce economically significant innovations.⁸⁶ Different groups also appear to have used the patent system in distinct ways: artisans and medium-skilled workers accounted for many patents of invention, whereas patents of importation were more often associated with commercially oriented and well-connected applicants and more frequently involved multiple applicants. These patterns suggest that skilled inventors generated many of the most valuable innovations, while commercially oriented actors used patents of importation to introduce and exploit technologies developed elsewhere.

[INSERT TABLE 8 HERE]

Placed in international comparative perspective (Table 8), the UKNL patent system shares many of the broad characteristics identified by recent cliometric studies of nineteenth-century European patent regimes, while also displaying several distinctive features.⁸⁷ As in Britain, France, Baden, the Italian states and Spain, the evidence suggests both that the patent system facilitated technology transfer and was broadly accessible to non-elite inventors, and that skilled patentees were disproportionately associated with economically valuable inventions.

Our findings therefore reinforce the emerging revisionist consensus that European patent systems were more inclusive than the ‘first-wave’ cliometric patent histories had suggested, despite relatively high fees and substantial administrative discretion.⁸⁸ However, the UKNL differs from several contemporary systems in important respects. Unlike studies of England, Prussia, and Württemberg, we find little evidence that administrative discretion translated into systematic discrimination against particular regions or social groups. And the availability of data on both successful and unsuccessful patent applications allows us to examine a question that other studies cannot address: the determinants of patent grant decisions.

⁸⁵ Compare with Billington, ‘Patenting behaviour’.

⁸⁶ Nuvolari, Tartari and Tranchero, ‘Patterns of innovation’; Nuvolari, Tortorici and Vasta, ‘British-French technology’.

⁸⁷ See Appendix B for a comparative synthesis of the cliometric literature on historical patent systems.

⁸⁸ Streb, ‘Inventing ideas’.

Finally, our findings reinforce Joel Mokyr's interpretation of the geography of industrialisation in the Low Countries.⁸⁹ More patents were granted in the Southern Netherlands, even after adjusting for population (see Figure 4). The South had more manufacturers as patentees and made greater use of patents of importation, consistent with technological catch-up through the adoption of technologies developed in France and England. In absolute terms, applicants from the Southern Netherlands also patented more than twice as many valuable innovations, even though, according to our indicator, a Northern patent was more likely to be valuable *ex post*.

But our findings also qualify Mokyr's argument that the Netherlands lacked institutional mechanisms for translating useful knowledge into industrial development.⁹⁰ The UKNL patent system represents one such mechanism. Rather than merely granting exclusive rights, it encouraged the domestic introduction of foreign technologies while tailoring patent protection to perceived technological and economic needs. Both the Kingdom of Belgium and the rump Kingdom of the Netherlands inherited this institutional framework, yet the former industrialised much more rapidly than the latter. The patent system was therefore not sufficient to generate industrialisation on its own, but it did provide an institutional mechanism through which foreign technologies could be identified, evaluated, and diffused in both successor states.

Taken together, these findings suggest that the UKNL patent system was more than a mechanism for allocating exclusive rights. It functioned as a policy instrument through which administrators sought to balance incentives for innovation against the costs of monopoly. Appendix F develops a simple conceptual model that formalises this interpretation by reverse-engineering the decision problem implied by the archival evidence and statistical analysis. The model provides an economic interpretation of the observed administrative practices and shows how the same underlying mechanisms can explain both the benefits of discretion and the subsequent divergence between Belgium and the Netherlands.

VI. CONCLUSION

Our quantitative analysis of all the UKNL's patent records reveals a patent application screening system that, despite granting administrators significant discretionary power to tailor the terms and conditions of each successful application, was not subject to abuse. Instead, discretion appears to have been exercised to optimise both the term and cost of patents on economic grounds, balancing the interests of inventors and broader society. These powers were applied consistently and non-arbitrarily; the system aligns closely with what Douglass North

⁸⁹ Mokyr, *Industrialization*.

⁹⁰ Mokyr, 'The Industrial Revolution and the Netherlands'.

might have classified as an open access order institution.⁹¹ The decline of the patent system in the Northern Netherlands following Belgian independence does not, in our view, indicate that the system was incapable of functioning equitably or efficiently. Notably, the same patent legislation continued to operate on both sides of the border after the dissolution of the UKNL.

In terms of outcomes, we find that the UKNL's patent system remained accessible to individuals outside the elite despite high fees; similar proportions of different social classes secured patents; and both technical skill and market orientation were important drivers of successful engagement with the system. At the same time, patent administrators systematically differentiated between patents of invention and patents of importation. Patents of importation were more frequently rejected and, when granted, were generally subject to shorter terms and were much less likely to receive fee discounts. They were also more commonly sought by commercially oriented applicants, whereas patents of invention were associated with technically skilled inventors.

Taken together, these differences suggest that administrators may have been pursuing what could cautiously be termed a technology or industrial policy, tailoring patent protection to the distinct economic functions of invention and technology transfer rather than applying inflexible uniform rules.⁹² This contrasts with the English case, where responsibility for patent screening was largely delegated to private parties and, as Billington and Lane recently argue, did not achieve the same balance between rewarding inventors and protecting the public interest.⁹³

All in all, we find that the broad discretion embedded in the UKNL's patent system did not result in unfair or inconsistent treatment. On the contrary, we argue that the system was administered in a manner that Acemoglu and Robinson might describe as characteristic of an inclusive economic institution.⁹⁴ This stands in contrast to van den Berg's interpretation of the broader integration of the UKNL's legal systems, which he depicts as an oppressive top-down imposition by Northern Netherlanders upon the Southern Netherlands.⁹⁵ In contrast to the inequitable treatment of the southern provinces with respect to political representation, fiscal policy, and religious freedoms, the UKNL's national innovation system was open and inclusive, and therefore unlikely to have contributed meaningfully as a motivating grievance for the Belgian Revolution of 1830.⁹⁶

⁹¹ North, Wallis and Weingast, *Violence*.

⁹² Compare with Wintle, *An economic and social history* (p. 136), who argues that contemporary economic thinking was insufficiently developed for there to have been an explicit economic policy, and that the state's financial and administrative capacity limited what government could realistically achieve.

⁹³ Billington and Lane, 'Success and failure'.

⁹⁴ Acemoglu and Robinson, *Why nations fail*.

⁹⁵ Van den Berg, 'De integratieve functie'.

⁹⁶ Kennedy, *A concise history*.

A further consideration, drawing on North's institutional framework, is that the effectiveness of discretion in the UKNL's patent system may have been contingent on its relatively limited scale.⁹⁷ With application volumes remaining modest, administrators were able to exercise case-by-case judgement without generating excessive transaction costs, enforcement problems, or scope for opportunism. After Belgian independence, however, rising application volumes in the southern Low Countries strained the administrative viability of this discretionary system. The Belgian system consequently shifted towards more formalised, rules-based procedures to preserve administrative capacity and predictability.⁹⁸

The rump Netherlands followed the opposite trajectory. Lower demand for patent protection, declining administrative investment, and weakening legal enforcement reduced both the use of the system and political support for its continuation, culminating in abolition in 1869.⁹⁹ These divergent post-UKNL trajectories suggest that discretionary patent institutions can function effectively at a small scale, but that expansion in volume and complexity tends to necessitate greater reliance on standardised rules, whereas institutional decline can instead produce a self-reinforcing cycle of neglect. This is consistent with North's emphasis on formal institutions as mechanisms for managing transaction costs.¹⁰⁰

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⁹⁷ Appendix F formalises this mechanism using an analytic narrative and extends it to explain the divergent post-1830 evolution of the Belgian and Dutch patent systems.

⁹⁸ Wagenaar, 'The patent system', in addition to discussion in Appendix F.

⁹⁹ Courts applying the same patent legislation reached different legal judgments in the 1840s, with a Belgian court upholding strong exclusive rights while Dutch courts ruled that copying for personal use was not prohibited, thereby undermining the enforceability of patents (Wagenaar, 'The patent system').

¹⁰⁰ North, *Institutions*.

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TABLES

Table 1: Definitions of variables used in statistical analysis

Variable	Unit	Definition
Patent of invention	Discrete	Applicant requests a patent of invention (1 = patent of invention application).
Patent of importation	Discrete	Applicant requests a patent for importing (introducing) a foreign invention (1 = patent of importation application).
Patent grant	Discrete	Patent is granted following examination of patent file (1 = granted).
Legal test failure	Discrete	Patent application is discontinued after a rejection that cites the law (1 = discontinued). Common reasons: (1) no clear description or drawing; (2) not in the correct language; (3) no patent lifespan or start date supplied for the foreign patent being imported; (4) invention not patentable subject matter.
Time to decision	Months	Number of months between date of first application and final decision.
Lifespan requested	Years	Term requested by the patent applicant in years.
Lifespan assigned	Years	Term assigned by patent administrators at the point of grant.
Difference in lifespan	Years	Difference in years between the term requested and the term assigned.
Lifespan reduced	Discrete	Term assigned at the point of patent grant is shorter than the term requested by the patent applicant (1 = smaller).
Difference in fees	Guilders	Difference between patent fees assigned and patent fees normally required by law (positive = cheaper).
Fees reduced	Discrete	Patent fees are discounted by 50 guilders or more from what is normally required by law. 50 guilders is the smallest fee discount given by officials.
Special conditions	Discrete	Patent is granted under special conditions, such as a compulsory licence clause or a shorter working clause (1 = conditions assigned).
Doorman indicator	Discrete	Whether the invention was published independently of the inventor, received a prize, or made a difference to technological history, as recorded by Doorman (1 = valuable patent).
Distance to capital	km (100s)	Distance the applicant must travel to the closest capital city (Brussels or The Hague), calculated using Google Maps. The UKNL alternated between its capitals every year. For each patent, the closer capital is used.
Multiple applicants	Discrete	Patent has multiple applicants (1 = more than one).
Experienced applicant	Discrete	One of the patent applicants has previously applied for a patent for a different invention (1 = has previously applied).
Industrial adviser	Discrete	Government has sought industry advice (1 = sought advice). Usually drawn from local Chambers of Commerce and Industry, but sometimes important manufacturers, such as John Cockerill or J. J. Huyttens Kerremans.

Continued on next page...

Variable	Unit	Definition
Technical adviser	Discrete	Government has sought technical advice (1 = sought advice). Usually a technical adviser employed by the ministry, but sometimes drawn from the Royal Academy of Science of Brussels, the Royal Institute of Science of Amsterdam, or other scientific societies or scientists.
Country of first applicant	Categorical	Country of residence of first-named applicant. Values are Northern UKNL (present-day Netherlands), Southern UKNL (present-day Belgium, Limburg and Luxembourg), or foreigners (usually France, German states or UK).
HISCLASS	Categorical	Variable based on the HISCLASS assigned to the first applicant. HISCLASS is usually divided into 12 classes. We collapse these into five class groups, in addition to an unknown category. See Appendix C for methodology.
Technology class	Categorical	Technology class, assigned from the title of the invention. 20 categories, including an unknown category. Based on the scheme used in Billington and Hanna, ‘That’s classified!’.
Steam-related technology	Discrete	Whether the invention relates to steam power (1 = steam-related).
Time	Categorical	Time bins of three-year windows, grouping patents together by year of first contact with the ministry. The first bin is a six-year window (1816–1821).

Notes: The distinction between dependent and explanatory variables is not made here; dependent variables in some regression specifications function as independent variables in others, depending on the question being asked.

Table 2: Univariate statistics for the full sample

Panel A: Full sample, patent applications of invention, and patent applications of importation

Variable	Full sample (n = 1,059)		Invention (n = 614)		Importation (n = 423)		Means test	
	<i>mean</i>	<i>sd</i>	<i>mean</i>	<i>sd</i>	<i>mean</i>	<i>sd</i>	<i>t-value</i>	
Patent grant (dummy)	0.482	0.500	0.560	0.497	0.400	0.490	5.292	***
Legal test failure (dummy)	0.220	0.414	0.205	0.404	0.229	0.421	-0.980	
Time to decision (months)	3.714	5.149	3.537	5.381	4.064	4.870	-1.688	*
Distance to capital (100 km)	0.905	1.402	0.495	0.599	1.524	1.941	-10.534	***
Multiple applicants (dummy)	0.111	0.315	0.098	0.297	0.137	0.344	-1.952	*
Experienced applicant (dummy)	0.191	0.393	0.202	0.402	0.177	0.382	1.044	
Industrial adviser (dummy)	0.038	0.191	0.018	0.133	0.069	0.253	-3.789	***
Technical adviser (dummy)	0.300	0.458	0.327	0.470	0.270	0.444	2.056	**
Steam-related technology (dummy)	0.069	0.253	0.047	0.212	0.095	0.293	-2.874	***
<i>HISCLASS:</i>								
Factory owners, nobility, landholders	0.183	0.387	0.138	0.346	0.253	0.435	-4.571	***
Higher professionals	0.076	0.266	0.060	0.238	0.102	0.303	-2.267	**
Merchants, clerks, lower professionals	0.129	0.336	0.122	0.328	0.139	0.347	-0.852	
Medium-skilled workers, business holders	0.211	0.408	0.280	0.449	0.118	0.323	6.943	***
Low-skilled workers	0.059	0.237	0.083	0.276	0.026	0.159	4.180	***
Class unknown	0.341	0.474	0.316	0.465	0.362	0.481	-1.606	

Notes: See Table 1 for variable definitions. Applications for patents of improvement are excluded; t-statistics assume unequal variance; significance levels reported as *** p<0.01, ** p<0.05, * p<0.1.

Panel B: Granted versus not granted applications for patents of invention and importation

Variable	Invention					Importation				
	Granted (n = 344)		Not granted (n = 270)		Means test	Granted (n = 169)		Not granted (n = 254)		Means test
	<i>Mean</i>	<i>sd</i>	<i>mean</i>	<i>sd</i>	<i>t-value</i>	<i>mean</i>	<i>sd</i>	<i>mean</i>	<i>sd</i>	<i>t-value</i>
Time to decision (months)	4.095	4.303	2.826	6.440	-2.785 ***	4.910	4.460	3.501	5.056	-3.016 ***
Distance to capital (100 km)	0.433	0.520	0.575	0.679	2.829 ***	1.322	1.512	1.659	2.173	1.888 *
Multiple applicants (dummy)	0.090	0.287	0.107	0.310	0.709	0.130	0.337	0.142	0.349	0.340
Experienced applicant (dummy)	0.224	0.417	0.174	0.380	-1.542	0.249	0.433	0.130	0.337	-3.004 ***
Industrial adviser (dummy)	0.020	0.141	0.015	0.121	-0.522	0.053	0.225	0.079	0.270	1.052
Technical adviser (dummy)	0.404	0.491	0.230	0.421	-4.731 ***	0.420	0.495	0.169	0.376	-5.600 ***
Steam-related technology (dummy)	0.058	0.234	0.033	0.180	-1.484	0.124	0.331	0.075	0.264	-1.629
<i>HISCLASS:</i>										
Factory owners, nobility, landholders	0.166	0.372	0.104	0.305	-2.266 **	0.337	0.474	0.197	0.398	-3.176 ***
Higher professionals	0.070	0.255	0.048	0.214	-1.140	0.071	0.258	0.122	0.328	1.787 *
Merchants, clerks, lower professionals	0.134	0.341	0.107	0.310	-0.999	0.195	0.398	0.102	0.304	-2.578 **
Medium-skilled workers, business holders	0.270	0.444	0.293	0.456	0.607	0.118	0.324	0.118	0.323	-0.007
Low-skilled workers	0.076	0.265	0.093	0.290	0.749	0.012	0.108	0.035	0.185	1.649 *
Class unknown	0.285	0.452	0.356	0.4780	1.859 *	0.266	0.443	0.425	0.495	3.444 ***

Notes: See Table 1 for variable definitions. Applications for patents of improvement are excluded; t-statistics assume unequal variance; significance levels reported as *** p<0.01, ** p<0.05, * p<0.1.

Panel C: Northern versus Southern UKNL applicants; and UKNL versus Foreign applicants

Variable	Northern versus Southern UKNL (all patents)					UKNL versus Foreign (patents of importation only)				
	Northern (n = 226)		Southern (n = 686)		Means test	UKNL (n = 286)		Foreign (n = 138)		Means test
	<i>Mean</i>	<i>sd</i>	<i>mean</i>	<i>sd</i>	<i>t-value</i>	<i>mean</i>	<i>sd</i>	<i>mean</i>	<i>sd</i>	<i>t-value</i>
Patent of invention (dummy)	0.748	0.435	0.636	0.482	3.271 ***					
Patent of importation (dummy)	0.217	0.413	0.344	0.475	-3.863 ***					
Patent grant (dummy)	0.442	0.498	0.515	0.500	-1.886 *	0.407	0.492	0.384	0.488	0.452
Legal test failure (dummy)	0.261	0.440	0.204	0.403	1.722 *	0.225	0.418	0.239	0.428	-0.331
Time to decision (months)	4.077	7.498	3.518	4.182	1.067 *	4.026	4.879	4.143	4.869	-0.231
Distance to capital (100 km)	0.602	0.484	0.424	0.467	4.809 ***	0.464	0.448	3.714	2.005	-18.816 ***
Multiple applicants (dummy)	0.128	0.335	0.105	0.307	0.928	0.151	0.359	0.109	0.312	1.239
Experienced applicant (dummy)	0.173	0.379	0.217	0.413	-1.502	0.218	0.413	0.094	0.293	3.528 ***
Industrial adviser (dummy)	0.027	0.161	0.038	0.191	-0.876	0.074	0.262	0.058	0.235	0.622
Technical adviser (dummy)	0.345	0.476	0.300	0.459	1.238	0.288	0.453	0.232	0.424	1.242
Steam-related technology (dummy)	0.049	0.216	0.070	0.255	-1.228	0.091	0.288	0.101	0.303	-0.330
<i>HISCLASS:</i>										
Factory owners, nobility, landholders	0.155	0.363	0.187	0.390	-1.119	0.270	0.445	0.217	0.414	1.200
Higher professionals	0.053	0.224	0.061	0.240	-0.464	0.056	0.231	0.196	0.398	-3.818 ***
Merchants, clerks, lower professionals	0.173	0.379	0.112	0.316	2.160 **	0.137	0.344	0.145	0.353	-0.223
Medium-skilled workers, business holders	0.217	0.413	0.235	0.424	-0.561	0.133	0.341	0.087	0.283	1.477
Low-skilled workers	0.084	0.278	0.063	0.243	1.034	0.035	0.184	0.007	0.085	2.125 **
Class unknown	0.319	0.467	0.343	0.475	-0.667	0.368	0.483	0.348	0.478	0.414

Notes: See Table 1 for variable definitions. Applications for patents of improvement are excluded; t-statistics assume unequal variance; significance levels reported as *** p<0.01, ** p<0.05, * p<0.1. Comparison between North and South is for the full sample; between the UKNL and foreign applicants is only for patents of importation.

Table 3: Univariate statistics for granted patents of invention and importation

Variable	Full sample			Invention			Importation			Means test	
	<i>N</i>	<i>mean</i>	<i>sd</i>	<i>N</i>	<i>mean</i>	<i>sd</i>	<i>N</i>	<i>mean</i>	<i>sd</i>	<i>t-value</i>	
Time to decision (months)	510	4.379	4.376	344	4.095	4.303	169	4.910	4.460	-2.092	**
Lifespan requested (years)	495	10.904	3.905	330	10.630	4.119	168	11.426	3.402	-2.363	**
Lifespan assigned (years)	510	8.854	3.117	344	8.997	3.230	169	8.525	2.857	1.557	
Difference in lifespan (years)	495	1.624	2.509	330	1.258	2.242	168	2.375	2.889	-4.343	***
Lifespan reduced (dummy)	495	0.311	0.463	330	0.245	0.431	168	0.440	0.498	-4.332	***
Difference in fees (guilders)	510	62.615	128.428	344	84.590	139.990	169	16.775	83.239	6.770	***
Fees reduced (dummy)	510	0.269	0.444	344	0.337	0.473	169	0.124	0.331	5.797	***
Special conditions (dummy)	510	0.090	0.287	344	0.029	0.168	169	0.213	0.411	-5.633	***
Doorman indicator (dummy)	510	0.120	0.325	344	0.102	0.303	169	0.160	0.367	-1.680	*
Distance to capital (100 km)	510	0.724	1.050	344	0.433	0.520	169	1.322	1.512	-7.374	***
Multiple patents (dummy)	510	0.104	0.305	344	0.090	0.287	169	0.130	0.337	-1.386	
Experienced applicant (dummy)	510	0.231	0.422	344	0.224	0.417	169	0.249	0.433	-0.573	
Industrial adviser (dummy)	510	0.031	0.174	344	0.020	0.141	169	0.053	0.225	-1.763	*
Technical adviser (dummy)	510	0.410	0.492	344	0.404	0.491	169	0.420	0.495	-0.377	
Steam-related technology (dummy)	510	0.080	0.272	344	0.058	0.234	169	0.124	0.331	-2.374	**
<i>HISCLASS:</i>											
Factory owners, nobility, landholders	510	0.224	0.417	344	0.166	0.372	169	0.337	0.474	-4.224	***
Higher professionals	510	0.069	0.253	344	0.070	0.255	169	0.071	0.258	0.147	
Merchants, clerks, lower professionals	510	0.155	0.362	344	0.134	0.341	169	0.195	0.398	-1.803	*
Medium-skilled workers, business holders	510	0.218	0.413	344	0.270	0.445	169	0.118	0.324	4.752	***
Low-skilled workers	510	0.055	0.228	344	0.076	0.265	169	0.012	0.108	3.825	***
Class unknown	510	0.280	0.450	344	0.285	0.452	169	0.266	0.443	0.326	

Notes: See Table 1 for variable definitions. Applications for patents of improvement are excluded; t-statistics assume unequal variance; significance levels reported as *** p<0.01, ** p<0.05, * p<0.1.

Table 4: Logistic regression of patent type (0 = invention; 1 = importation)

Variable	Full sample				Granted patents			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Patent grant (dummy)	-0.643*** (0.146)	-0.730*** (0.163)	-0.820*** (0.171)	-0.854*** (0.167)				
Difference in lifespan (10log)					0.161*** (0.049)	0.163*** (0.054)	0.181*** (0.057)	0.172*** (0.054)
Fees reduced (dummy)					-1.195*** (0.315)	-1.053*** (0.322)	-0.844** (0.344)	-0.648** (0.310)
Doorman indicator (dummy)					0.529 (0.357)	0.798** (0.362)	0.694* (0.398)	0.729* (0.400)
Time to decision (months)	0.018 (0.018)	0.019 (0.016)	0.023 (0.019)	0.029 (0.019)	0.020 (0.027)	0.029 (0.026)	0.048* (0.027)	0.049* (0.025)
Distance to capital (10log)	0.174*** (0.042)	0.002 (0.048)	-0.016 (0.050)	-0.016 (0.047)	0.197*** (0.067)	0.057 (0.072)	0.013 (0.080)	0.010 (0.076)
Multiple applicants (dummy)	0.332 (0.217)	0.392 (0.250)	0.480* (0.269)	0.554** (0.259)	0.078 (0.327)	0.327 (0.378)	0.460 (0.435)	0.596 (0.445)
Experienced applicant (dummy)	-0.124 (0.179)	0.064 (0.195)	-0.013 (0.200)	0.026 (0.194)	0.226 (0.264)	0.344 (0.299)	0.277 (0.302)	0.314 (0.270)
Industrial adviser (dummy)	1.187*** (0.391)	1.254*** (0.412)	1.350*** (0.444)	1.507*** (0.404)	0.994 (0.614)	1.134* (0.639)	1.210 (0.854)	1.035 (0.726)
Technical adviser (dummy)	-0.266 (0.163)	-0.102 (0.182)	-0.136 (0.187)	-0.112 (0.177)	-0.064 (0.241)	0.167 (0.267)	0.233 (0.271)	0.227 (0.249)
Steam-related technology (dummy)				0.842*** (0.286)				0.748** (0.352)
<i>Country of first applicant:</i>								
Northern UKNL		reference	reference	reference		reference	reference	reference
Southern UKNL		0.730*** (0.215)	0.739*** (0.224)	0.741*** (0.212)		0.857** (0.341)	0.979*** (0.369)	0.738** (0.333)
Foreign		4.293*** (0.419)	4.278*** (0.436)	4.207*** (0.436)		4.125*** (0.696)	4.349*** (0.702)	4.275*** (0.711)
<i>HISCLASS:</i>								
Factory owners, nobility, landholders			reference	reference			reference	reference
Higher professionals			-0.672** (0.339)	-0.753** (0.337)			-1.208* (0.630)	-1.218** (0.605)
Merchants, clerks, lower professionals			-0.367 (0.280)	-0.444* (0.265)			-0.078 (0.380)	-0.072 (0.347)
Medium-skilled workers, business holders			-1.493*** (0.258)	-1.541*** (0.249)			-1.702*** (0.393)	-1.419*** (0.373)
Low-skilled workers			-1.775*** (0.405)	-1.740*** (0.403)			-2.206*** (0.840)	-2.058*** (0.763)
Class unknown			-0.570** (0.228)	-0.617*** (0.218)			-0.908** (0.353)	-0.636* (0.339)
Constant	-0.543* (0.325)	-1.921*** (0.410)	-1.184*** (0.448)	-0.590** (0.293)	0.281 (0.500)	-0.926 (0.611)	-0.414 (0.683)	-1.011** (0.490)
Technology fixed effects	Yes	Yes	Yes	No	Yes	Yes	Yes	No
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,024	1,024	1,024	1,024	475	475	475	475
Patents of importation	420	420	420	420	165	165	165	165
Pseudo R-squared	0.086	0.233	0.267	0.251	0.151	0.268	0.315	0.282
AUC	0.694	0.790	0.819	0.807	0.758	0.823	0.850	0.828

Notes: See Table 1 for variable definitions. Reported coefficients are log-transformed odds ratios. Robust standard errors in parentheses; significance levels reported as *** p<0.01, ** p<0.05, * p<0.1.

Table 5: Logistic regression of granted patents (1 = granted)

Variable	Invention			Importation		
	(1)	(2)	(3)	(4)	(5)	(6)
Time to decision (months)	0.006 (0.024)	0.011 (0.026)	0.018 (0.036)	0.047 (0.030)	0.034 (0.032)	0.023 (0.028)
Distance to capital (10log)	-0.136* (0.071)	-0.160** (0.072)	-0.127* (0.068)	1.515*** (0.322)	1.484*** (0.329)	0.988*** (0.285)
Multiple applicants (dummy)	-0.217 (0.378)	-0.191 (0.383)	-0.079 (0.349)	0.744** (0.350)	0.557 (0.381)	0.677** (0.343)
Experienced applicant (dummy)	-0.210 (0.275)	-0.315 (0.280)	-0.299 (0.276)	-0.178* (0.097)	-0.236** (0.101)	-0.114 (0.087)
Industrial adviser (dummy)	0.273 (0.932)	0.300 (0.946)	0.526 (0.896)	-0.113 (0.300)	-0.189 (0.330)	-0.184 (0.309)
Technical adviser (dummy)	0.197 (0.236)	0.162 (0.238)	0.141 (0.225)	-1.131** (0.498)	-1.051* (0.539)	-0.828* (0.441)
Steam-related technology (dummy)			0.734 (0.623)			0.498 (0.406)
<i>Country of first applicant:</i>						
Northern UKNL	<i>reference</i>		<i>reference</i>	<i>reference</i>		<i>reference</i>
Southern UKNL	-0.072 (0.268)	-0.039 (0.267)	0.220 (0.252)	-0.480 (0.445)	-0.599 (0.451)	-0.281 (0.426)
Foreign	0.015 (0.814)	0.102 (0.880)	0.203 (0.876)	-0.210 (0.450)	-0.184 (0.460)	0.110 (0.446)
<i>HISCLASS:</i>						
Factory owners, nobility, landholders		<i>reference</i>	<i>reference</i>		<i>reference</i>	<i>reference</i>
Higher professionals		-0.356 (0.542)	-0.471 (0.510)		-1.154** (0.513)	-0.949** (0.482)
Merchants, clerks, lower professionals		-0.201 (0.428)	-0.224 (0.414)		-0.017 (0.443)	0.037 (0.397)
Medium-skilled workers, business holders		-0.571 (0.372)	-0.676* (0.364)		-0.376 (0.479)	-0.463 (0.423)
Low-skilled workers		-0.900* (0.489)	-0.891* (0.470)		-0.132 (0.882)	-1.000 (0.889)
Class unknown		-0.621 (0.392)	-0.610 (0.374)		-1.081*** (0.361)	-0.901*** (0.333)
Constant	-0.828* (0.468)	-0.449 (0.535)	0.146 (0.396)	0.253 (0.740)	0.947 (0.831)	0.125 (0.627)
Technology fixed effects	Yes	Yes	No	Yes	Yes	No
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	488	488	488	315	315	326
No. of positives	339	339	339	169	169	169
Pseudo R-squared	0.078	0.086	0.059	0.167	0.197	0.103
AUC	0.690	0.698	0.667	0.762	0.782	0.710

Notes: See Table 1 for variable definitions. Reported coefficients are log-transformed odds ratios. Robust standard errors in parentheses; significance levels reported as *** p<0.01, ** p<0.05, * p<0.1.

Table 6: Regression of patent fees, lifespan reductions, and time to decision for granted patents

Variable	Fees reduced		Lifespan reduced		Time to decision	
	Invention	Importation	Invention	Importation	Invention	Importation
	Logit (1)	Logit (2)	Logit (3)	Logit (4)	OLS (5)	OLS (6)
Time to decision (months)	0.078** (0.036)	-0.020 (0.086)	0.029 (0.032)	0.050 (0.044)		
Lifespan requested (years)	-0.070* (0.038)	-0.065 (0.101)			0.179*** (0.059)	0.055 (0.109)
Doorman indicator (dummy)	0.597 (0.404)	1.679* (0.956)	-0.037 (0.448)	-0.812 (0.636)	0.976 (0.787)	1.408 (1.069)
Distance to capital (10log)	-0.232** (0.097)	0.164 (0.164)	-0.039 (0.096)	-0.040 (0.141)	0.107 (0.155)	0.171 (0.260)
Multiple applicants (dummy)	-1.786*** (0.594)	-0.301 (1.082)	0.217 (0.464)	0.152 (0.568)	-1.426* (0.819)	-0.642 (1.072)
Experienced applicant (dummy)	-0.751** (0.376)	-0.322 (0.880)	-0.496 (0.377)	-0.098 (0.425)	-0.615 (0.593)	-2.454*** (0.850)
Industrial adviser (dummy)	-0.104 (0.746)	2.194* (1.258)	-0.924 (0.950)	0.850 (0.885)	-0.556 (1.618)	0.651 (1.613)
Technical adviser (dummy)	0.404 (0.312)	-2.032* (1.174)	-0.618* (0.336)	0.173 (0.414)	-0.053 (0.509)	1.406* (0.779)
<i>Country of first applicant:</i>						
Northern UKNL	<i>reference</i>	<i>reference</i>	<i>reference</i>	<i>reference</i>	<i>reference</i>	<i>reference</i>
Southern UKNL	-0.013 (0.445)	1.905* (1.151)	-0.382 (0.381)	0.926 (0.653)	-0.424 (0.641)	-0.019 (1.160)
Foreign	1.791* (1.060)		1.254 (1.395)	0.564 (0.625)	-1.092 (2.207)	-0.710 (1.210)
<i>HISCLASS:</i>						
Factory owners, nobility, landholders	<i>reference</i>	<i>reference</i>	<i>reference</i>	<i>reference</i>	<i>reference</i>	<i>reference</i>
Higher professionals	0.963* (0.552)	-1.968* (1.119)	0.309 (0.599)	-0.219 (0.758)	-1.402 (1.076)	0.084 (1.418)
Merchants, clerks, lower professionals	0.478 (0.566)	-0.278 (0.898)	-0.382 (0.525)	0.509 (0.515)	0.391 (0.867)	-0.381 (1.036)
Medium-skilled workers, business holders	1.528*** (0.442)	0.433 (0.970)	-0.184 (0.487)	0.351 (0.675)	2.192*** (0.737)	-1.759 (1.214)
Low-skilled workers	1.294** (0.619)		0.039 (0.645)	0.300 (1.431)	1.533 (1.054)	-0.391 (3.438)
Class unknown	0.203 (0.518)	-0.533 (0.788)	0.868* (0.455)	0.046 (0.489)	1.704** (0.779)	-0.636 (0.955)
Constant	-1.195 (0.864)	-2.432 (2.396)	-1.508* (0.850)	-0.180 (0.995)	0.370 (1.396)	1.246 (2.427)
Technology fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	330	121	323	159	330	168
No. of positives	114	21	81	74	n/a	n/a
(Pseudo) R-squared	0.235	0.327	0.138	0.113	0.203	0.271
AUC	0.815	0.865	0.751	0.725	n/a	n/a

Notes: See Table 1 for variable definitions. Dependent variable in (1) and (2) is a dummy variable, where 1 = applicant received a fee discount of 50 guilders or more; dependent variable in (3) and (4) is a dummy variable, where 1 = the applicant was assigned a shorter patent lifespan than requested. Reported coefficients are log-transformed odds ratios. Robust standard errors in parentheses; significance levels reported as *** p<0.01, ** p<0.05, * p<0.1.

Table 7: Logistic regression of *ex post* patent value

Variable	Invention	Importation
	(1)	(2)
Time to decision (months)	0.046 (0.042)	0.057 (0.066)
Lifespan requested (years)	0.027 (0.043)	-0.168* (0.090)
Distance to capital (10log)	0.128 (0.142)	0.110 (0.166)
Multiple applicants (dummy)	0.416 (0.514)	-2.750*** (0.923)
Experienced applicant (dummy)	-0.589 (0.512)	-2.282** (0.963)
Industrial adviser (dummy)	0.023 (0.962)	0.125 (1.306)
Technical adviser (dummy)	-0.124 (0.445)	0.258 (0.695)
<i>Country of first applicant:</i>		
Northern UKNL	<i>reference</i>	<i>reference</i>
Southern UKNL	-0.722 (0.565)	-0.646 (0.914)
Foreign	0.165 (1.148)	-1.847* (0.998)
<i>HISCLASS:</i>		
Factory owners, nobility, landholders	<i>reference</i>	<i>reference</i>
Higher professionals	1.609** (0.746)	-0.234 (0.948)
Merchants, clerks, lower professionals	-1.197 (0.837)	-2.387* (1.221)
Medium-skilled workers, business holders	0.173 (0.646)	-0.499 (0.827)
Low-skilled workers	-0.451 (0.995)	
Class unknown	-0.333 (0.607)	-1.892** (0.900)
Constant	-1.968 (1.259)	-0.795 (1.736)
Technology fixed effects	Yes	Yes
Time fixed effects	Yes	Yes
Observations	302	155
No. of high-value patents	34	27
Pseudo R-squared	0.154	0.396
AUC	0.768	0.900

Notes: See Table 1 for variable definitions. Dependent variable is a dummy variable, where 1 = patent has a Doorman indicator. Reported coefficients are log-transformed odds ratios. Robust standard errors in parentheses; significance levels reported as *** p<0.01, ** p<0.05, * p<0.1.

Table 8: Comparison of findings across cliometric studies of historical European patent systems

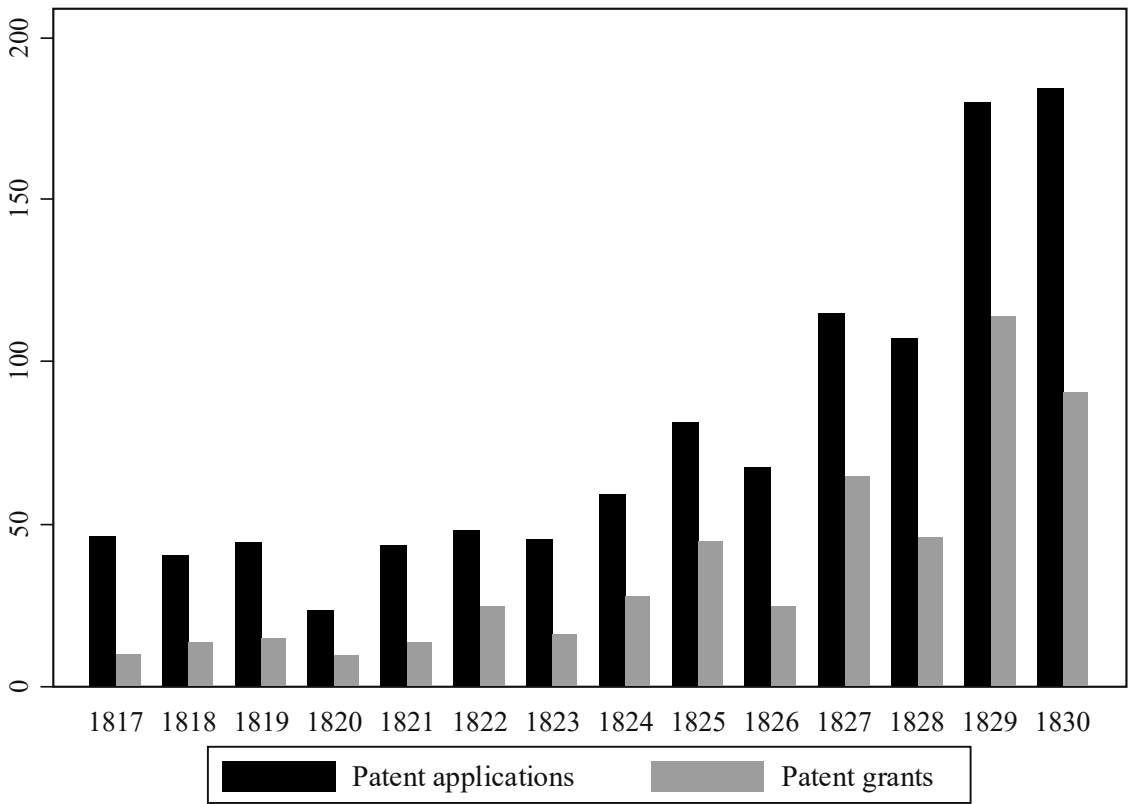
Study site:	UK	France	Baden	Württemberg	Prussia	Italian states	Spain	Sweden	UKNL
Study window:	1700–1841	1791–1844	1844–1877	1844–1868	1845–1877	1855–1872	1820–1930	1840–1914	1817–1830
Study author(s):	Billington	Nuvolari, Tortorici and Vasta	Donges and Selgert	Lehmann-Hasemeyer and Streb	Donges and Selgert	Nuvolari and Vasta	Sáiz	Berger and Prawitz	This study
<i>(1) Patents and technology transfer</i>									
Patents are an important means of technology transfer	Yes	Yes	Yes	Yes	--	Yes	Yes	--	Yes
Patents of importation differed fundamentally	--	No	--	--	--	--	Yes	--	Yes
<i>(2) Accessibility and discrimination within the patent system</i>									
Patent system is accessible to non-elite patentees	Yes	Yes	Yes	Yes	Yes	Yes	--	No	Yes
Domestic regions are treated differently	--	No	No	--	--	No	--	--	No
Foreigners are treated differently	--	No	No	Yes	Yes	No	--	--	Yes
<i>(3) Demand- and supply-side drivers of patenting</i>									
Timing and geography of patenting mirrors Industrial Revolution	Yes	Yes	Yes	Yes	No	Yes	--	--	Yes
Skilled patentees are the most frequent users of the patent system	Yes	Yes	Yes	--	Yes	--	--	Yes	Yes
Patentees obtain patents in jurisdictions with market opportunities	Yes	--	Yes	Yes	Yes	Yes	--	--	Yes
<i>(4) The measurement of patent value</i>									
Valuable patents are patented in more jurisdictions	Yes	Yes	--	Yes	--	Yes	--	Yes	Yes
Skilled patentees patent more valuable inventions	Yes	Yes	--	--	--	--	--	Yes	Yes

Notes: In bold we note the conclusions drawn from quantitative analysis. Non-bold are based on literature or qualitative reasoning by these authors. Dashed line (--) indicate hypotheses that are not discussed or tested in that publication.

Sources: UK: Billington, ‘What explains’; France: Nuvolari, Tortorici and Vasta, ‘British-French technology’; Baden: Donges and Selgert, ‘Technology transfer’; Württemberg: Lehmann-Hasemeyer and Streb, ‘Discrimination’; Prussia: Donges and Selgert, ‘The social background’; Italian states: Nuvolari and Vasta, ‘Patenting the Risorgimento’; Spain: Sáiz, ‘Did patents’; Sweden: Berger and Prawitz, ‘Inventors’.

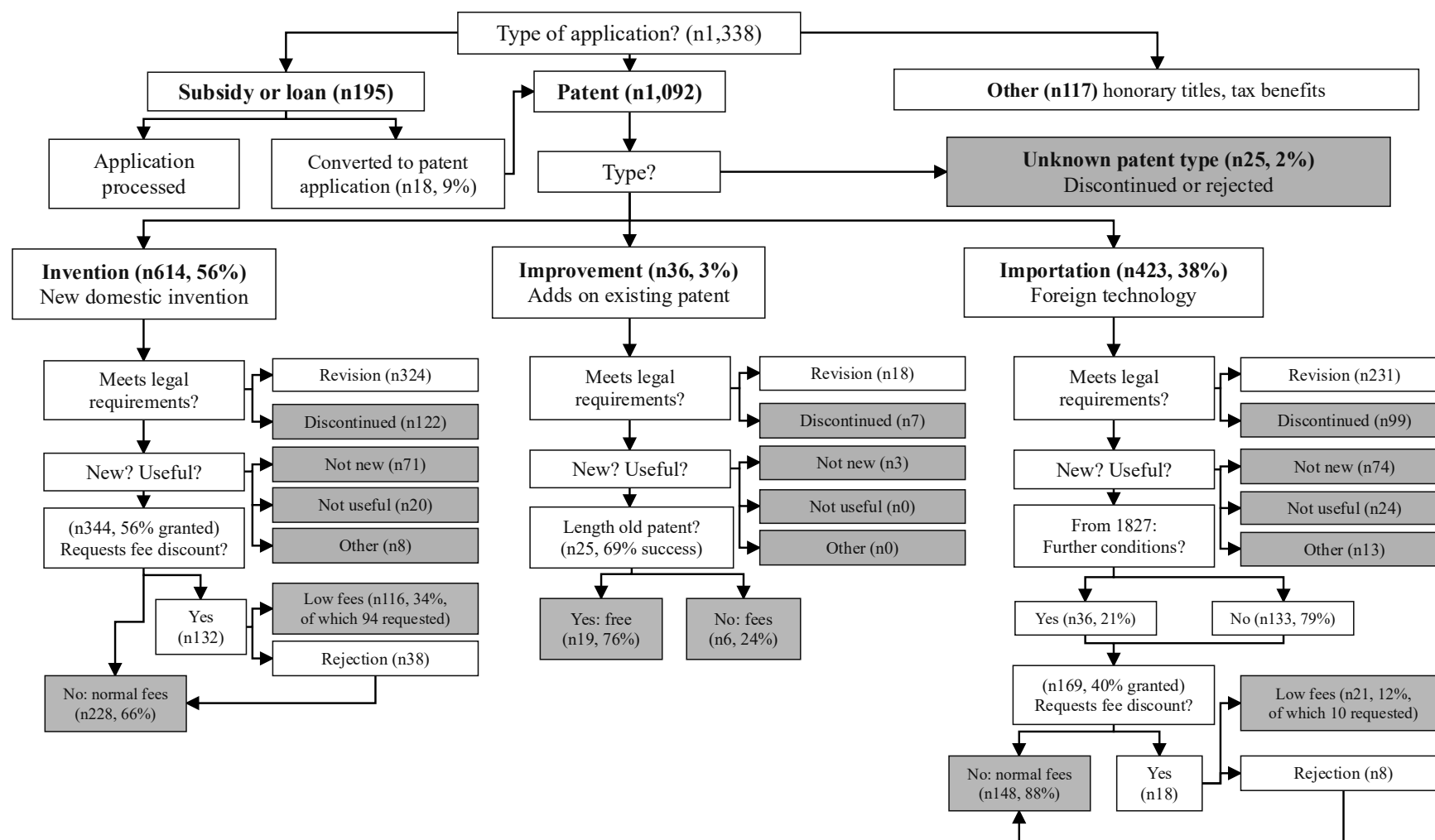
FIGURES

Figure 1: Patent applications and patent grants, all types, 1817–1830



Sources: Authors’ calculations using UKNL patent database.

Figure 2: Flowchart depicting the UKNL patent system, 1817–1830



Notes: Orange shading signifies rejected or withdrawn patents; green shading represents all granted patents.

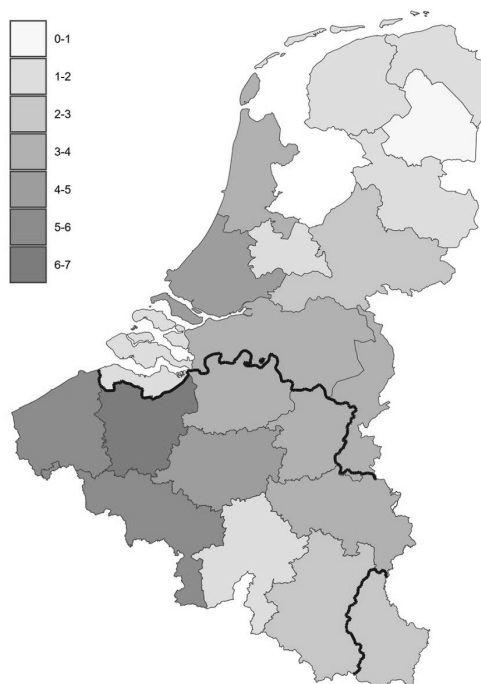
Sources: Authors' depiction of the patent system as part of the UKNL's wider national innovation system, as described in Appendix A.

Figure 3: Maps of the 19 provinces of the UKNL, 1819

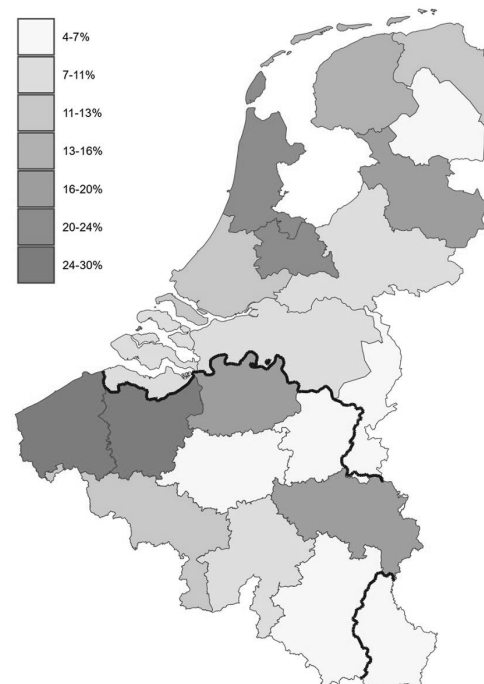
(a) Border locations and province names



(b) Total population, in 100,000s



(c) Industry and mining employment, in per cent

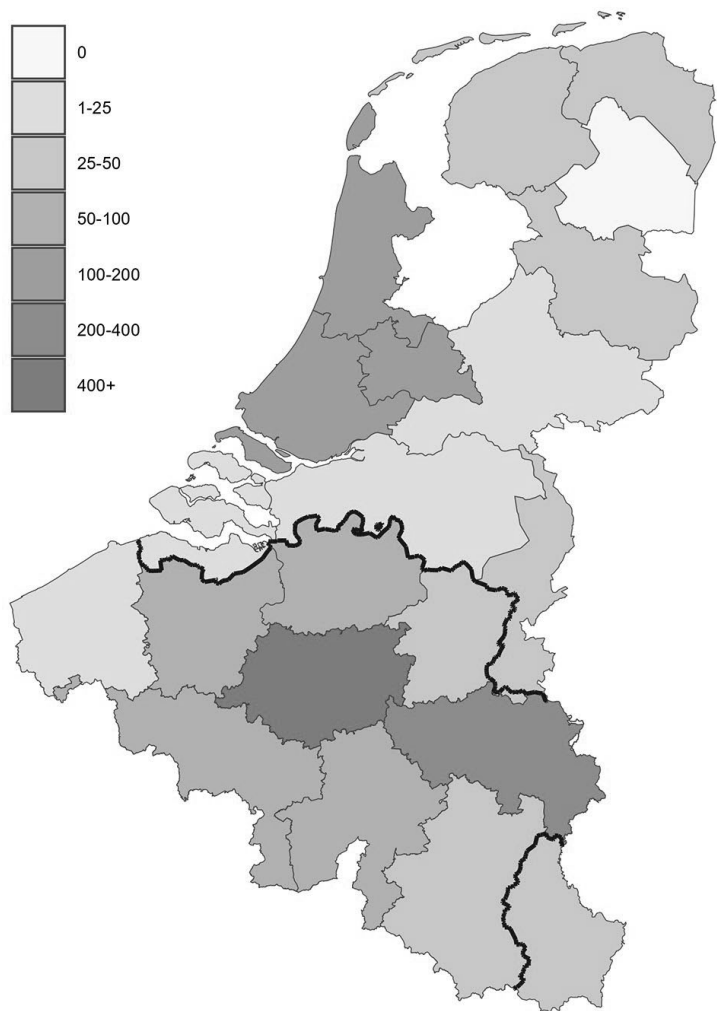


Notes: Map depicts the 19 provinces of the UKNL, superimposing the international borders following the Treaty of London in 1839 (in blue). NHOL, ZHOL, UTRE, ZEEL, FRIE, GRON, DRENT, OVER, GELD and NBRA constitute 10 of the 11 provinces that make up the UKNL's successor state of the Netherlands. WVLA, OVLA, ANTW, ZBRA, HAIN, NAMU and LIEG are 7 of the 9 provinces that form the successor state of Belgium. LIMB was partitioned between these two successor states, while LUXE was partitioned between Belgium and Luxembourg. We include both halves of both provinces in our definition of the Southern UKNL in our analysis.

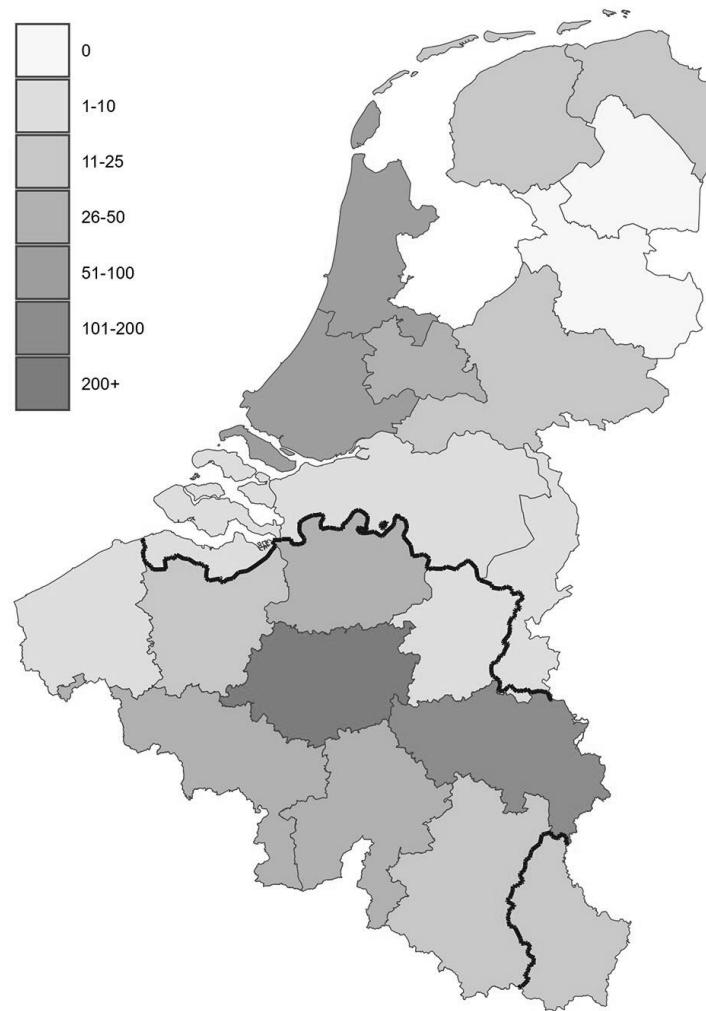
Sources: Population data for 1819 from Commissie voor de Statistiek, *Opgave der bevolking*; employment data from the 1819 industrial census extracted by Philips and Buyst, 'Regional industrialization'.

Figure 4: Maps of patenting activity across the UKNL, 1817–1830

(a) Patent of invention applications, per capita



(b) Patent of invention grants, per capita



Sources: Authors' calculations using patent database and population data from Commissie voor de Statistiek, *Opgave der bevolking*.

ONLINE APPENDICES

Patently Peculiar: Patents and Innovation in the United Kingdom of the Netherlands

A DESIGN OF THE UKNL PATENT SYSTEM

This appendix provides a detailed account of the administrative procedures governing patent applications in the UKNL, drawing directly on archival case files to reconstruct how applications were assessed, revised, and ultimately granted or rejected. Building on earlier work by Wagenaar, it complements the empirical analysis of our article's main text by documenting the sequential stages of patent examination and highlighting where procedures diverged across patent types.¹ Particular attention is paid to the role of administrative discretion, the use of external expertise, and the mechanisms through which patent fees, conditions and terms were adjusted in response to both applicant characteristics and perceived economic utility.

The administrative processes for different patent types began similarly. As indicated in Figure 2 (in the main text), all applications underwent an initial formal test to ensure they met minimum legal requirements. The most common grounds for rejection at this stage was an inadequate description (or drawing) of the invention, a description in the wrong language, or a lack of information submitted on the original patent for a patent of importation.² Most applicants were allowed to revise and resubmit their applications if they failed at the first hurdle; only rarely were patents immediately rejected for falling outside patentable subject matter.³

Complete specifications then underwent substantive examination for novelty, utility, and other considerations.⁴ Unlike registration-based patent systems, this stage gave civil servants considerable discretion in determining whether an application should proceed and, if so, under what conditions. Contemporary discussions surrounding the drafting of the 1817 patent law indicate that ministers regarded the discretionary powers of civil servants as a safeguard against

¹Wagenaar, 'Rules'; Wagenaar, 'The patent system'.

²The official languages for the UKNL were Dutch and French. From January 1823 the UKNL government compelled applicants from the Dutch-speaking provinces to address the government in Dutch. This included South-Brabant, which as a province was Dutch-speaking but where the city of Brussels was largely Francophone. Those in French-speaking provinces, such as Namur or Liège, could continue to address the government in French. See Vosters and Janssens, 'Willems taalpolitiek', for a discussion of this language policy.

³Pharmaceuticals were not patentable because patent protection was deemed to interfere with state control of medicines. Chemical and food recipes were likewise excluded because officials considered patents based on the precise composition of ingredients impossible to enforce. Rather than granting rights that could not be enforced, and which might therefore be abused to signal quality where none existed, they refused to patent such inventions altogether. See 'Report to the King on a request for Rumfordsch Soeppoeder', NL-HaNA, 2.04.01, 4106, 21-1-1821, 39/112 [all translations of document descriptions by authors].

⁴Other considerations included the potential impact of an invention on state tax revenues and whether it posed a danger to public safety. One example of the latter was an application for an umbrella designed to conceal a rifle.

insignificant inventions, arguing that patent protection should be reserved for inventions that genuinely merited exclusive rights.⁵

Surviving patent files reveal how administrators understood the purpose of this discretion in practice: balancing incentives for invention against the wider costs of monopoly. In recommending patent terms, they argued that inventors should be rewarded sufficiently to encourage innovation, but that exclusive rights should be limited to the shortest duration compatible with that objective. As one report in June 1817 explained:

If, as the law has established, it is fair to ensure that the author of a discovery enjoys the benefits of that discovery; if it is good to do so to encourage new research and motivate those who achieve successful results to benefit their country, it is no less useful, to reconcile general interest with individual interest, that this enjoyment be limited to the shortest possible term; for it is nothing more for the person who obtains it than the privilege of placing the public in their dependence and making them reliant on them for the objects of the invention: it also follows that if these products are truly useful, only a small part of the public can benefit from them, both because they are put on the market in too small quantities and because they are sold at too high a price. These considerations will always serve in the future as a basis for the proposals that I will have the honor to make to Your Majesty regarding the duration of the patents that I will submit for its approval. These are the reasons that lead me to propose to His Majesty to limit the attached patent to ten years for Messrs. Hartsinck & Co and Blanckenay, whose secret is so important that it seems certain that during this period they will be able to reap the greatest benefits, benefits that will serve them sufficiently as a reward.⁶

This report is important because, unlike much of the archival record, it states the principles underlying administrative discretion explicitly rather than revealing them only indirectly through individual decisions. Rather than treating patent duration as fixed, administrators viewed it as an instrument to be adjusted so that inventors received a sufficient reward while monopoly protection remained no more extensive than necessary. Implementing these principles required information. Administrators could consult external advisers on an invention's utility and novelty, although they were often confident enough to reach their own judgements.⁷ The extent of consultation varied across applications and patent types, reflecting differences in technical complexity and the information available to the administration.

Administrative practice diverged by patent type at the point at which patents were ready to be granted. Applicants for a patent of invention or importation could hope to obtain a reduction in their patent fees; more than a third of the applications for a patent of invention requested a patent

⁵Letter of the Minister of the Interior to the Commissioner General for Education, Arts and Sciences', 15-11-1816, No. 5; and 'Report of the Council of State concerning the Draft Patent Act of 1817', 13-12-1816, both in NL-HaNA, 2.04.01, 4039.

⁶'Report to the King', NL-HaNA, 2.04.01, 4040, 17-6-1817, no. 1172 (Hartsinck & Blanckenay file) [translation by the authors].

⁷See, e.g., the way that civil servants dismissed applications for technically impossible perpetual motion devices, discussed in Wagenaar, 'The patent system'.

fee discount. This request set in motion an additional investigation: the Ministry of Education would inquire after the applicant's personal wealth, moral standing, and whether they had a caring responsibility to dependents. In the archives, we find reports drafted by applicants' local mayors, who gathered information from the police, local business owners and the applicants' employees to assess their financial circumstances, reputation and family responsibilities.⁸

Alongside applicants' personal circumstances, patent administrators would also consider their estimate of the invention's utility. For example, if the value of the invention was deemed to be high, administrators would communicate to the applicant that they should try to find an investor instead of seeking a patent.⁹ If they questioned whether the invention would work at all, administrators preferred to keep patent fees high to discourage frivolous patents.¹⁰ The procedure was much simpler for patents of improvement. These top-up patent rights were generally granted free of charge, provided applicants requested the same duration as the original patent. Patent fees also served a broader policy purpose. Under Article IX of the 1817 patent law, the fee proceeds were paid into a ring-fenced fund 'for the promotion of arts and industry', part-financing the patent system's administration, in addition to fee discounts, premiums and rewards for inventors.

The administration applied these principles differently to patents of importation. Rather than treating them as equivalent to patents of invention, administrators first considered whether exclusive rights were necessary to encourage the domestic introduction of foreign technologies. Fee discounts were therefore rarely granted. Indeed, they often questioned whether such patents should be issued at all, and under what terms. Where they judged commercial incentives to be sufficient, they shortened patent terms or imposed additional conditions instead.¹¹ This policy reached full maturity around 1827, after two major scandals involving patents of importation.¹² Chambers of Commerce were henceforth asked to assess the feasibility of importation and the acceptability of the patent to local industry.¹³ If supportive, these bodies typically recommended

⁸See, e.g., 'Letter to the Governor of Antwerp on a request for a free patent by van Campen', NL-HaNA, 2.04.01, 4145, 19-12-1822, 2210.

⁹Report to the King on the request by van Campen', NL-HaNA, 2.04.01, 2.04.01, 4148, 1-2-1823, 45/218.

¹⁰It is advisable to multiply the grant of patents without payment of duties as little as possible, from the perspective that the payment of duties gives the government a means that is simple and offends no-one to prevent countless patent requests for useless and pipe-dream inventions' [translation by authors]. 'Report on request L. Embach to obtain a free patent', NL-HaNA, 2.04.01, 2.04.01, 4256, 29-10-1825, 137A.

¹¹See, e.g., 'Report to the King on Thomas Jackson for a patent of importation of an invention by Brown', NL-HaNA, 2.04.01, 4218, 1-11-1824, no 133A.

¹²In 1825–6 and 1827–8, two related scandals arose in the Southern Netherlands over patents of importation obtained by proxies: Magnan for John Collier, and Kockx for Albert (representing Jones). Both concerned machines for finishing draperies linked to French entrepreneurs. Rather than construct the machines locally, Magnan and Kockx acted as commercial agents importing for their patrons, prompting textile manufacturers successfully to challenge the practice. Magnan lost all his patents, while Kockx had to transfer his to a local manufacturer under a new open-licence clause. These cases made the government more cautious about such patents. See 'Report to the King on resistance to Magnan's patents', NL-HaNA, 2.04.01, 4267, 10-1-1826, 114A; and 'Letter to the governor of Liège on Kockx', NL-HaNA, 2.04.01, 4414, 29-7-1828, 25F.

¹³This test had a standard formulation: (1) Is this invention yet unknown and unused in this kingdom? (2) Is it to be expected that this invention will be imported without a patent of importation? (3) Is it useful to stimulate this importation with a patent? And: (4) under what conditions? See 'Letter to the governor of East-Flanders in Ghent on Warin's patent request for an invention by J. Nicholson', NL-HaNA, 2.04.01, 4350, 14-8-1827, 17F.

Table A1: All foreign patent grantees which had additional conditions imposed on their patents

Name	Country	Original patentee	Patent description	Year	Additional condition
Bent	England	Raddatz	Steam engine	1827	Short working clause
Ternaux	France	Delcourt	Machine for preparing flax	1829	Short working clause
Berg	England	None	Processes to filter / refine sugar	1829	Open license
Camusat	France	Pelletan	Process for making soda	1829	Open license
Chasselon	France	Noverre	Machine for kneading bread	1829	Open license
Chasselon	France	Maisonneuve	Machine for kneading bread	1829	Open license
Duplessis	France	Souchon	Replacing indigo for fabrics	1829	Open license
Davis	England	Gardner and Herbert	Machine to prepare draperies	1830	Short working clause

Notes: We exclude one UKNL national living in France who had a shorter working clause imposed, as he was a patent agent representing an English inhabitant whose patent concerned an imported salt-making technology.

adding an open licence clause and reducing the requested patent term. This led to a four-pronged treatment of patents of importation: rejection of easily accessible inventions; reduction of patent lengths to acceptable levels; open license clauses; and stricter working clauses.¹⁴

These policies translated into a distinctive pattern of patent conditions. Across the full body of 153 patents of importation granted, only 8 had additional conditions.¹⁵ Table A1 reports the titles of patents on which conditions were imposed. All relate to industries with substantial domestic economic activity. What is also evident from this list is the difference in names between the original patentees and the patent applicant in the UKNL; no patent was requested under the same name as the original patent.¹⁶

B CLIOMETRIC STUDIES OF PATENT SYSTEMS

Cliometric research on patent systems has developed into a substantial field within economic history, using large-scale patent data to analyse innovation, technology transfer, and institutional performance during the First and Second Industrial Revolutions. Early work treated patents primarily as indicators of inventive activity, but more recent studies emphasise the institutional features of patent systems and their interaction with economic incentives, social structures, and international technology flows. The literature can be organised around four interrelated themes: (1) patents and technology transfer; (2) accessibility and discrimination within the patent system; (3) demand- and supply-side drivers of patenting; and (4) the measurement of patent value. Table 8 (in the main text) summarises the findings of the most recent cliometric articles on eight nineteenth-century European patent systems along these four dimensions, alongside the

¹⁴In the flowchart in Figure 2, ‘further conditions’ refers to patents subject to either an open licence or a stricter working clause.

¹⁵There were also 10 patents of invention that had additional conditions imposed upon them. Because they represent a very marginal fraction of this patent type, we did not incorporate this step in the flowchart in Figure 2.

¹⁶Besides entrepreneurs seeking patents of importation for inventions patented abroad by others, this pattern probably reflects an idiosyncratic rule in both the French and UKNL systems: obtaining a patent for the same invention in another country nullified the original patent (Wagenaar, ‘Rules’). Because authorities could verify this only when patentees used their own names, they had an incentive to employ proxies or agents.

conclusions of our own research on the patent system of the UKNL.¹⁷

A central concern in cliometric patent research is the role of patent institutions in facilitating cross-border technology transfer. Historical patent systems frequently granted monopoly rights not only for new inventions, but also for the domestic introduction of foreign technologies. Such patents of importation or introduction were widespread in Europe during the eighteenth and early nineteenth centuries and reflected a conception of novelty defined relative to the polity rather than globally. Comparative studies document considerable institutional variation. England permitted patents for imported technologies from an early date, before the emergence of modern novelty requirements. France's 1791 patent law, by contrast, explicitly distinguished patents of invention from patents of importation.¹⁸ This latter model was subsequently adopted in other jurisdictions influenced by French law, including Spain, several Italian states, and parts of Germany.¹⁹ By contrast, the US departed from this tradition by restricting patents to the 'first and true inventor' from 1790 onwards.²⁰

Empirically assessing the impact of patent systems on technology transfer is challenging, as it requires comparing patented and unpatented innovations. Cliometric studies have approached this problem in two main ways. One strand exploits cross-jurisdictional variation in patent regimes, showing that differences in patent use were associated with patterns of technological diffusion, as in Baden or pre-unification Italy.²¹ Another strand analyses multi-jurisdiction filings, interpreting foreign patenting as evidence of expected market potential and international diffusion. Studies of Britain, France, Spain, and colonial settings consistently find that valuable technologies were more likely to be patented across multiple jurisdictions.²² However, Nuvolari and co-authors find that Britain was an exception; patents by foreign applicants were, in general, of lower quality than those by domestic applicants.²³

Despite the institutional prominence of patents of importation, few quantitative studies distinguish them systematically from invention patents. While the French system recognised patents of importation as a distinct intellectual property right, Nuvolari and co-authors combine the two in their statistical analysis.²⁴ To our knowledge, Sáiz's work remains the notable exception, showing for nineteenth-century Spain that importation patents tended to be shorter and more costly, while being accessible to both domestic and foreign applicants.²⁵

¹⁷These eight cases are: (1) the UK (England, Scotland and Ireland); (2) France; (3) Baden; (4) Württemberg; (5) Prussia; (6) the pre-unification Italian states; (7) Spain; and (8) Sweden.

¹⁸For France, see Baudry, 'Une histoire'; Nuvolari, Tortorici and Vasta, 'British–French technology transfer'; Spain: Pretel, 'Institutionalising patents'; Sáiz, 'Did patents'.

¹⁹For Italy, see Armengaud, 'Guide de l'inventeur'; Tolhausen, 'A synopsis'; Nuvolari and Vasta, 'Independent invention'; and for Germany: Donges and Selgert, 'Technology transfer'.

²⁰Khan, 'The democratization'.

²¹Nuvolari and Vasta, 'Patenting the Risorgimento'; Donges and Selgert, 'Technology transfer'.

²²Britain: Bottomley, 'The British patent system'; Billington, 'What explains patenting'; France: Nuvolari, Tortorici and Vasta, 'British–French technology transfer'; Spain: Sáiz, 'Did patents'; and colonial settings: Graham, 'Patents of invention'.

²³Nuvolari, Tartari and Tranchero, 'Patterns of innovation'; Nuvolari and Tartari, 'Bennet Woodcroft'.

²⁴Nuvolari, Tortorici and Vasta, 'British–French technology transfer'.

²⁵Sáiz, 'Did patents'.

A second major theme concerns accessibility and discrimination within patent systems. The debate was shaped by Sokoloff and Khan's influential claim that the early US patent system democratised invention by enabling widespread participation by artisans, rural inventors, and individuals outside traditional elites.²⁶ Khan extended this argument by contrasting the allegedly inclusive US system with what she characterised as elitist European systems, citing high fees, weak legal security, and slow publication.²⁷ Subsequent research has substantially qualified this interpretation. Comparative studies show that artisans and non-elite inventors patented extensively in Britain, France, and Italy, despite higher costs and weaker enforcement than in the US.²⁸ Other work emphasises that European systems relied on self-selection through fees and informal screening rather than on formal exclusion.²⁹

Beyond social class, scholars have examined other dimensions of exclusion. Gender has received particular attention, with evidence that marriage laws and legal incapacity sharply constrained female patenting in the US and elsewhere.³⁰ Race has been studied mainly in the US context, where Cook documents how racial violence suppressed African American patenting.³¹ Nationality has also emerged as an important axis of discrimination. Many countries imposed explicit or implicit barriers on foreign patentees, including citizenship requirements, higher fees, compulsory licensing, and administrative delays. Studies of nineteenth-century Prussia and Württemberg, and early twentieth-century Germany and Japan show that such barriers were often applied selectively and informally, even when formal treaty obligations existed.³²

A third strand of the literature situates patenting within broader debates on the drivers of technological change during the First Industrial Revolution. While scholars agree that many important innovations occurred outside the patent system, patent data have been used to test competing demand- and supply-side explanations.³³ On the supply side, studies consistently find that skilled individuals – artisans, engineers, and scientifically trained inventors – dominated patenting and were associated with more valuable patents in the cases of Britain, France, Sweden, and the German states.³⁴ On the demand side, research shows that patenting responded strongly to market incentives: inventors were more likely to patent where markets existed for their

²⁶Sokoloff and Khan, 'The democratization'.

²⁷Khan, 'The democratization'.

²⁸Britain: Billington, 'What explains patenting'; France: Nuvolari, Tortorici and Vasta, 'British–French technology transfer'; and Italy: Nuvolari and Vasta, 'Independent invention'; Nuvolari and Vasta, 'The geography'; Nuvolari and Vasta, 'Patenting the Risorgimento'.

²⁹Bottomley, 'Patents'; Baudry, 'Une histoire'; Galvez-Behar, 'The patent system'.

³⁰Khan, 'Married women'; Khan, 'Not for ornament'; Khan, 'Invisible women'; Khan, 'Designing women'; Khan, 'A new way'.

³¹Cook, 'Violence'.

³²Prussia: Donges and Selgert, 'The social background'; Württemberg: Lehmann-Hasemeyer and Streb, 'Discrimination'; early twentieth-century Germany: Richter and Streb, 'Catching-up'; and Japan: Kotabe, 'A comparative study'; Nicholas, 'Independent invention'.

³³MacLeod, 'Inventing'; MacLeod, 'Patents for invention'; Mokyr, 'Intellectual property rights'; Moser, 'Patents and innovation: evidence'; Moser, 'Patents and innovation in economic history'; Cox, 'Patent disclosure'.

³⁴Donges and Selgert, 'Technology transfer'; Billington, 'What explains patenting'; Nuvolari, Tortorici and Vasta, 'British–French technology transfer'; Berger and Prawitz, 'Inventors'.

innovations, and more valuable patents were disproportionately extended across jurisdictions.³⁵

Patent grant rates generally track the timing of industrialisation across countries, although notable exceptions exist, such as Prussia's highly restrictive regime.³⁶ The causal relationship between patenting and industrialisation remains contested. While patents clearly responded to technological change, their role as a driver rather than a reflection of industrialisation is less certain.³⁷ The literature nonetheless converges on the view that patenting was particularly attractive in industries where secrecy was difficult and infringement was easily detected.³⁸

Finally, cliometric research has devoted substantial attention to measuring patent value. Scholars emphasise the limitations of patent statistics: many inventions were not patented, patent scope varied widely, and patents were often used strategically for signalling or branding rather than protection.³⁹ To address these issues, innovation economists and economic historians employ a range of proxies, including renewal behaviour, foreign filings, expert assessments of specifications, citation analysis, inventor biographies, and alternative indicators such as productivity growth, world fair exhibits, and prizes.⁴⁰

Taken together, this literature portrays historical patent systems as heterogeneous institutions whose effects depended critically on legal design, administrative practice, and economic context. While consensus has emerged on the importance of patents for technology diffusion and on the broad accessibility of many European systems, significant disagreement remains over the extent of discrimination, the relative importance of demand and supply factors, and the interpretation of patent value. Largely absent are studies like ours that examine the correlates of successful patent applications, reflecting the limited survival and digitisation of the necessary archival sources. These unresolved issues motivate our context-specific analysis of the determinants of patent success and administrative discretion within the UKNL patent system.

³⁵Bottomley, 'Patenting in England'; Nuvolari and Vasta, 'Patenting the Risorgimento'; Nuvolari, Tartari and Tranchero, 'Patterns of innovation'.

³⁶Donges and Selgert, 'The social background'.

³⁷Greasley and Oxley, 'Patenting'; MacLeod and Nuvolari, 'Inventive activities'.

³⁸Moser, 'How do patent laws'.

³⁹MacLeod, 'Inventing'; Griliches, 'Patent statistics'; Streb, 'The cliometric study'.

⁴⁰Renewal behaviour: Schankerman and Pakes, 'Estimates'; foreign filings: Cantwell, 'Technological innovation'; expert assessments of specifications: MacLeod, Tann et al., 'Evaluating inventive activity'; citation analysis: Nuvolari and Tartari, 'Bennet Woodcroft'; inventor biographies: Khan and Sokoloff, 'Schemes'; productivity growth: Moser, 'Patents and innovation: evidence'; world fair exhibits: Moser, 'How do patent laws'; and prizes: Brunt, Lerner and Nicholas, 'Inducement prizes'.

C INVENTOR OCCUPATIONS CLASSIFICATION

To examine the social background of patent applicants and the accessibility of the patent system, we classify occupations using the Historical International Social Class Scheme (HISCLASS), developed by Marco van Leeuwen and Ineke Maas on the basis of their Historical International Standard Classification of Occupations (HISCO).⁴¹ The schema combines occupation codes with status indicators that can also be found in historical data (student, owner, boss, apprentice, etc.), and relationship indicators when the occupation mentioned in the historical source is held by a family member rather than the individual in question. HISCLASS distinguishes between 12 class categories that are subdivided into four skill levels (high, medium, low and unskilled), as well as distinctions showing manual/non-manual, supervision/non-supervision and primary sector/non-primary sector.⁴²

Table C1 shows the results of this exercise for our dataset. The last two columns report data from Billington's study of English patent grantees between 1700 and 1841 for comparison.⁴³ The social composition of patentees is broadly similar across the two countries, with rank correlations of 0.833 (Spearman) and 0.674 (Kendall's Tau-b). The principal difference is that manufacturers, nobles, military officers, and property owners (HISCLASS 1) form the largest group of patentees in England, whereas artisans and medium-skilled workers (HISCLASS 6–7) dominate in the UKNL. England also has a larger share of higher professionals, probably reflecting a greater presence of engineers, while the UKNL has relatively more traders, lower professionals, and clerical workers (HISCLASS 3–5).

The interpretation of these differences is less straightforward. They may reflect the greater accessibility of the UKNL patent system, where patent fees could be adjusted to applicants' means, or they may simply capture differences in economic structure. Britain's more advanced industrialisation would naturally be expected to produce larger shares of manufacturers, managers, and engineers among patentees. Since patenting expanded over time in both countries, with most observations drawn from the later years of each series, the two explanations are difficult to disentangle. Even so, both patent systems were dominated by skilled individuals. In the UKNL, artisans (HISCLASS 6) and higher professionals (HISCLASS 2) together accounted for more than two-fifths of patentees.

We re-categorised selected HISCLASS categories (Table C2) where the standard classification was not well suited to the occupational structure of the early nineteenth century. Categories 1 and 2 remain unchanged. Categories 3–5 were merged to group merchants and clerks as non-manual

⁴¹van Leeuwen, Maas and Miles, 'HISCO'; van Leeuwen and Maas, 'HISCLASS'.

⁴²1 = Higher managers (high skill level), 2 = Higher professionals (high), 3 = Lower managers (medium skill level), 4 = Lower professionals and clerical and sales personnel (medium), 5 = Lower clerical and sales personnel (low), 6 = Foremen (medium), 7 = Medium-skilled workers (medium), 8 = Farmers and fishermen (medium), 9 = Lower-skilled workers (low), 10 = Lower-skilled farm workers (low), 11 = Unskilled workers (unskilled), 12 = Unskilled farm workers (unskilled).

⁴³Billington, 'What explains patenting'. We do not include the French data of Nuvolari, Tartari and Tranchero, 'Patterns of innovation', because they do not closely follow the HISCO/HISCLASS codification system.

Table C1: Patents by HISCLASS categories

HISCLASS		UKNL patents				UKNL patentees				England patentees	
No.	Label	Applications		Grants		Applicants		Grantees		Grantees	
		<i>N</i>	%	<i>N</i>	%	<i>N</i>	%	<i>N</i>	%	<i>N</i>	%
1	Higher managers	199	18.22%	118	22.18%	127	15.28%	72	18.60%	1,881	35.75%
2	Higher professionals	85	7.78%	38	7.14%	60	7.22%	28	7.24%	800	15.21%
3	Lower managers	43	3.94%	28	5.26%	20	2.41%	11	2.84%	17	0.32%
4	Lower professionals	92	8.42%	51	9.59%	77	9.27%	43	11.11%	719	13.67%
5	Lower clerical, sales	8	0.73%	5	0.94%	8	0.96%	5	1.29%	51	0.97%
6	Foremen	35	3.21%	15	2.82%	33	3.97%	15	3.88%	25	0.48%
7	Medium-skilled workers	193	17.67%	101	18.98%	143	17.21%	78	20.16%	1,213	23.06%
8	Farmers, fishermen	5	0.46%	1	0.19%	4	0.48%	1	0.26%	45	0.86%
9	Low-skilled workers	60	5.49%	27	5.08%	55	6.62%	24	6.20%	486	9.24%
11	Unskilled workers	5	0.46%	2	0.38%	5	0.60%	2	0.52%	24	0.46%
-1	Unknown	367	33.61%	146	27.44%	299	35.98%	108	27.91%	0	0.00%
Total		1,092		532		831		387		5,261	

Notes: The division of British patents across HISCLASS is from Billington, ‘What explains patenting’. For comparability with his data, we report both patent and patentee counts for the UKNL.

workers. Categories 6–8 were combined into an artisan/medium-skilled worker/farmer group to avoid arbitrary distinctions based on artisans’ self-identification as masters. Low-skilled and unskilled workers were likewise merged because of the small number of unskilled workers.

While HISCO has been valuable for historians, it has limitations for this period. Professionalisation was still developing, and occupational titles were less standardised. For example, ‘engineer’ in the UKNL did not yet apply to mechanics, complicating the categorisation of individuals identifying as *mécanicien*, *werktuigkundige*, *mechanicus*, or ‘mechanic.’ Some installed machinery, while others designed machines, requiring different skill classifications. In England, ‘mechanical engineers’ already existed, leading to an upward bias in skill classification for foreign mechanics compared to UKNL counterparts.

Additionally, occupation data often fail to distinguish workers from proprietors. A brewer (employee) and a brewery owner are both classified as medium-skilled (7), with owners assigned one level higher if their status is known. Similarly, a master carpenter is placed under 6 and a regular carpenter under 7 – though guilds had already been abolished, making such distinctions less relevant.

Another challenge is the historical role of traders and merchants, which was more significant in the nineteenth century. Wholesalers are classified as 3 (lower managers) and retailers as 4 (lower professionals and sales personnel), though their wealth and social class often matched those of manufacturers. The Dutch term *koopman* covers both wholesalers and retailers, but HISCLASS classifies all as retailers, creating a downward bias. In contrast, French distinguishes between *négociant* (wholesaler) and *marchand* (retailer), leading to potentially inconsistent classifications between Dutch- and French-speaking merchants.

In their use of HISCO/HISCLASS, different studies have tried to resolve issues in their own

Table C2: Recoding of HISCLASS categories

Category label	HISCLASS	Represented professions (English translation)	<i>N</i>	%
Factory owners, nobility, landholders	1	Manufacturer, director, military officer, owner, baron, esquire	199	18.22%
Higher professionals	2	Engineer, medical doctor, chemist, pharmacist, priest, professor, lawyer, government adviser	85	7.78%
Merchant, clerks, lower professionals	3, 4, 5	Merchant, broker, shop keeper, book seller, lower engineer, teacher, pharmacy assistant	143	13.10%
Medium-skilled workers, business holders	6, 7, 8	Mechanic, watchmaker, smith, carpenter, tailor, lithographer, printer, music instrument maker, instrument maker, baker	233	21.34%
Low-skilled workers	9, 10, 11, 12	Barber, pencil maker, glove maker, salt maker, soap maker, worker	65	5.95%
Unknown	-1	Unknown, inventor, unemployed, pensioned military officer, student, widow	367	33.61%
Total			1,092	100%

Notes: Categories adapted from van Leeuwen and Maas, ‘HISCLASS’.

ways. While Billington and Berger and Prawitz have more faithfully followed the HISCLASS codification system, Nuvolari et al. have tried to solve some of the issues mentioned above by adapting the HISCLASS system.⁴⁴ They add two categories to signify ownership: 0-1 and 0-2, where 0-1 contains small proprietors such as traders/merchants, shopkeepers and jewellers, and 0-2 large proprietors such as manufacturers, owners and rentiers, while also not coding the HISCO/HISCLASS categories very faithfully.⁴⁵ Since Billington uses HISCLASS to test for the relationship between skill and patenting, he reduces the number of categories in his econometric analysis by using the system’s four skill levels, thus mitigating some of the problems of the system. Nuvolari et al. do not actually use the full HISCLASS system in their econometrics, except class 2 for testing for scientists/engineers and class 6 for testing for artisans.

We code occupations as faithfully as possible into HISCO categories, following the HISCO guide for job titles in French and Dutch, the HISCO occupation descriptions, the HISCLASS codifications for status and relation, and sometimes upgrading individuals to manufacturers when other historical source material shows their wealth or the size of their enterprise.⁴⁶ As discussed, we have recombined the HISCLASS categories to compensate for the weaknesses in the coding.

⁴⁴Billington, ‘What explains patenting’; Berger and Prawitz, ‘Inventors’; Nuvolari, Tortorici and Vasta, ‘British–French technology transfer’.

⁴⁵For example, Nuvolari, Tortorici and Vasta, ‘British–French technology transfer’, translate *fabricants* as large manufacturers and allocate them to HISCLASS categories 0–2 (large proprietors). HISCO, however, notes that in France a *fabricant* was more likely to be a small producer, whereas in Belgium the term could indeed refer to a manufacturer. They also classify chemists among the lower professionals (4), while HISCO/HISCLASS places them among higher professionals (2), and redefine category 6 as artisans, despite HISCLASS reserving it for medium-skilled manual workers with supervisory responsibilities, such as foremen and master artisans.

⁴⁶For example, musical-instrument maker Sax (the father of the inventor of the saxophone) was promoted from musical-instrument maker (7) to manufacturer (1) when it became clear from other source material that he led a workshop of 100 employees.

D PATENT TECHNOLOGY CLASSIFICATION

Patent classifications can be constructed in three different ways: according to the nature of the technical property, the industry of origin, or the industry of use. Each approach serves a different analytical purpose.⁴⁷ Carrying out our own classification is necessary because patents in the UKNL were not organised in a manner that facilitates definition, retrieval, or comparison. We classified patent titles adapted from the methodology developed by Stephen Billington and Alan Hanna.⁴⁸ This appendix sets out our classification procedure in detail, including the criteria applied and the steps followed in assigning patents to classes.

Owing to historical circumstances surrounding the Belgian revolution, the UKNL never published its patents, despite this being stipulated by law, nor were patent application numbers sufficiently large to necessitate a formal retrieval system. This helps to explain the absence of any official classification. Even had such a system existed, it would not necessarily be appropriate for the purposes of the present research. As Billington and Hanna argue, reliance on historical patent classifications ties analysis to the historical contingencies that shaped those taxonomies, which are not always aligned with research questions. While economic analyses are primarily interested in classifications based on industry of use, patent authorities classify patents by technology or industry of origin. Patent classification systems are rarely stable over time, as authorities have revised them in response to changes in society, the economy, and technology. Reliance on historical national classifications also impedes cross-country and intertemporal comparability.

To address these issues, Billington and Hanna developed a machine-learning algorithm based on patent titles. The method assumes that patents in similar fields employ similar vocabularies and therefore groups titles according to textual proximity. It then assigns class labels to these groups, drawing on a predefined taxonomy. In their application, Billington and Hanna surveyed both historical patent classifications and those developed in the economic and technological history literatures, deriving a taxonomy of twenty standard patent classes based on industry of use, including categories such as ‘agriculture’, ‘apparel’, ‘chemicals’, and ‘machinery’.

Compared with the hand-coded classifications commonly used in the economic history literature, this machine-learning approach offers greater efficiency as well as increased consistency, transparency, and replicability. Because the original tool was designed for the English language, we adapted it to the specific characteristics of Dutch, particularly its extensive use of compound words. For example, a water mill (*watermolen*) would not otherwise be linked to a grain mill (*graanmolen*) without decomposing the compound. We modified the algorithm accordingly.

An example is patent no. 35: ‘*Invoering distilleerwerktuigen tot eene voortdurende distillatie*’ (importation of distillation machines for a continuous distillation). To make sure it would link with patent no. 115 ‘*Distilleertoestel*’ (distillation apparatus), we needed to make sure the software would link *Distilleertoestel* to *distilleerwerktuig* and *distillatie*, while not include

⁴⁷Griliches, ‘Patent statistics’.

⁴⁸Billington and Hanna, ‘That’s classified!’

meaningless words such as *een* (a) and *tot* (for). We needed to cut compounds and reduce words to their stem, transforming patent no. 35 into: ‘*invoer distill werktuig voortduur distill*’ (import distill machine continue distill). To reduce a loss of meaning, as Dutch compounds are more than just an addition of their parts, we introduced a duplication mechanism, thus both including ‘*distilleerwerktuig*’ and ‘*distill werktuig*’ for the machine learning combinations.

To address the practical problems posed by stemming and compound words, we imported a Dutch-language stemmer and a comprehensive Dutch vocabulary, supplemented by vocabulary derived from the patent titles themselves.⁴⁹ To prepare the data, we manually standardised spelling across titles and removed residual French terms that would otherwise interfere with the analysis. We also added descriptive information in square brackets where official titles were overly terse. For example, patent no. 240, ‘*Jagtgeweer van de soort genaamd à percussion*’, was expanded to ‘*Jagtgeweer van de soort genaamd à percussion [Index: uitvinding van een middel ter voorkoming van ongelukken door het gebruik van vuurwapenen]*’ (Hunting rifle of the type called à percussion [Index: invention of a means to prevent accidents through the use of firearms]).

We further refined the stemmer manually to unify variant spellings and related terms. For example, we consolidated *destilleer* and *distilleer* under *destilleer*, and reduced words such as *fabriek* (factory), *fabrikaten* (factory output), *fabrikage*, and *fabriceeren* to *fabriceren* (factory production). The stemmer was applied twice, once before compound splitting and once afterwards, to ensure that split components were also stemmed consistently.

For compound splitting, we relied primarily on vocabulary-based recognition, ideally identifying both components of a compound as valid words. This was not always feasible. To improve performance, we introduced rules to prevent implausible splits, such as rejecting splits that produced components without vowels, of only a single letter, or containing consonant combinations that do not occur at the beginning or end of Dutch words (for example, *gsg*). We also implemented a rule allowing for linking letters such as *-n-* or *-s-*. Ultimately, a combination of coding and repeated rounds of manual correction proved necessary to achieve reliable splits and stems.

A second important preparatory step involved removing words that lack invention-specific meaning and would therefore generate spurious links. These included numbers, evaluative adjectives (such as ‘simple’, ‘difficult’, or ‘large’), common verbs (‘do’, ‘help’, ‘give’), and adverbs and prepositions (‘before’, ‘up’, ‘down’). We began with the stop-word list used by Billington and Hanna, translated into Dutch, and subsequently refined it to maximise meaningful connections. Removing too few words increases noise, while removing too many risks eliminating informative signals. Particularly difficult decisions arose with polysemous words. We excluded terms such as *stelsel* (‘system’), *behandelen* (‘treat’), and *stoffen* (‘textiles’, ‘ingredients’, or ‘matter’), while retaining words such as *dier* (‘of her’, but also ‘animal’), *vol* (‘full’, but also a textile process), and *gemak* (‘ease of use’, but also ‘toilet’), based on their observed effects on classification

⁴⁹Bird, Klein and Loper, ‘Natural language processing’.

outcomes.

Once frequency combinations were generated, the machine-learning tool produced a pre-defined number of ranked word lists. We then assigned a class to each list, enabling the algorithm to determine which word combinations corresponded to which classes. Using this input, the tool compared each patent title to the most closely aligned word lists and reported first-, second-, and third-best fits. Because the taxonomy comprises twenty classes and the algorithm produces more than twenty-word lists, the top-ranked matches may sometimes correspond to the same class.

This approach inevitably makes classification dependent on the composition of the corpus. Some inventions plausibly span multiple categories. An ‘improved water mill for grinding grain’, for example, combines power (‘water’), machinery (‘mill’), and output (‘grain’), and could therefore be classified under power, machinery, or food. Similarly, an invention describing a kettle for a steam engine on a ship could plausibly fall under power, machinery, or transportation. The assigned classification will depend on the relative prevalence of related patents within the corpus. Whether such biases are problematic depends on the research question, but users of the data must be aware of them. Machine-learning approaches do not eliminate deeper conceptual challenges in classification; their advantage lies in improved consistency, transparency, and replicability.

The method performs best in fields where titles contain frequently used words with unambiguous meanings, such as ‘weaving’ or ‘smithing’. It is less accurate for categories such as food and health, where titles often combine very generic verbs (‘make’, ‘prepare’, ‘produce’) with relatively rare identifying nouns (‘wine’, ‘bread’). As a result, food-related terms such as ‘wine’, ‘bread’, ‘cheese’ and ‘butter’ are not necessarily linked. Exceptions include ‘salt’, ‘alcohol’, and ‘sugar’, which were already industrially processed and sufficiently common to form coherent clusters. Similar issues arise for health-related terms, where specific diseases, body parts, or hygienic processes (such as ventilation) occur too infrequently to be reliably linked.⁵⁰

Table D1 summarises patenting activity across technology classes and reports associated outcomes, including grant rates, fee reductions, and the requested and granted length of patents. The classes in the table are ordered by the number of granted patents. It shows that patenting activity was unevenly distributed across sectors. Given these differences in both the volume of applications and their outcomes across technological fields, it is necessary to control for technology class in the regression analysis to account for sector-specific variation in patenting behaviour and administrative decisions.

Figure D1 provides a complementary visualisation of the patterns reported in Table D1. It compares the rank order of technology classes at the application and grant stages separately for patents of invention and importation, while arrow thickness represents the grant rate. The figure shows that the technological composition of granted patents did not simply reproduce

⁵⁰A related problem concerns inventions described using euphemistic language. We found that the wide range of euphemisms used for toilets – such as *gemak*, *secreet*, and *stoelgang* – were not linked by the algorithm. To improve accuracy in these limited subfields, we manually corrected the classification of the affected patents.

that of applications: some classes rose or fell in rank, while grant rates varied substantially across both technology classes and patent types. These differences provide further evidence that administrative outcomes were shaped partly by the technological field and whether an application concerned invention or importation, reinforcing the need to control for technology class and patent type in the regression analysis.

Transportation recorded the largest number of applications (153), followed by textiles (143) and food-related technologies (130). These sectors also produced a large number of granted patents, indicating strong inventive activity in areas linked to industrial production and infrastructure. A second group of sectors exhibited moderate levels of patenting activity, including power (85 applications), utility technologies (78), commodities (74), construction (72), and paper (47). Other fields, such as machinery (43), agriculture (39), metallurgy (33), and apparel (33), recorded fewer applications but in some cases relatively high success rates. Finally, more specialised sectors – including chemicals, military technologies, mining, and health-related technologies – accounted for comparatively small numbers of applications.

Grant rates varied considerably across technological fields. Mining exhibited the highest success rate (78%), followed by paper (73%), machinery (69%), and metallurgy (68%), while power (36%), chemicals (42%), and apparel (43%) recorded lower approval rates. In most sectors, patents of invention were more likely to be granted than patents of importation, and in some cases – notably mining and health – no importation patents were granted.

The incidence of fee reductions also differed across sectors. Among patents of invention, fee reductions were relatively common in apparel, chemicals, instruments, and paper, but less frequent in sectors such as textiles, machinery, and construction. These figures reflect the share of fee discount requests that were granted by the patent administration, indicating that administrators exercised discretion when determining the financial conditions under which patents were issued. Fee reductions for patents of importation were comparatively rare.

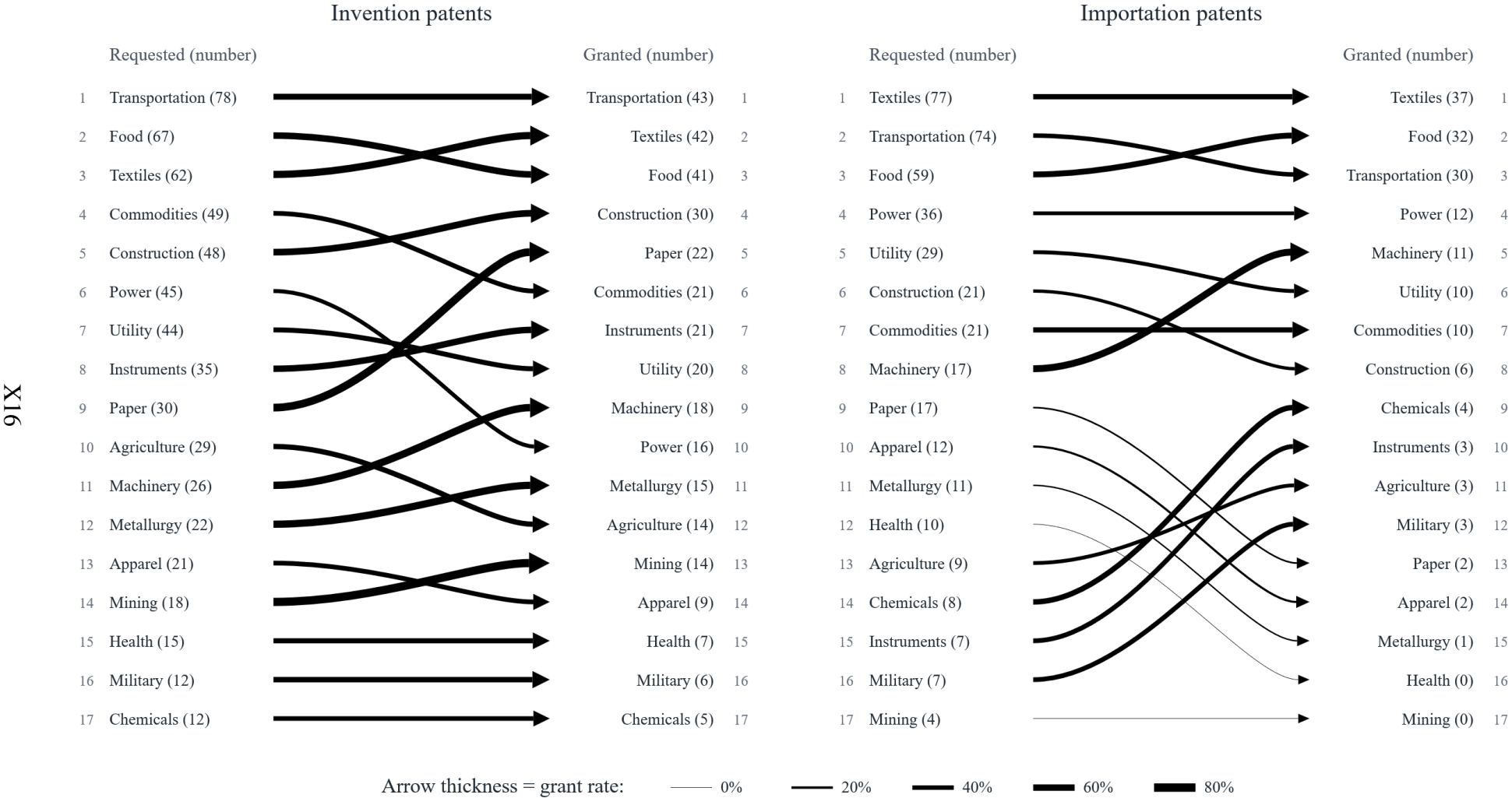
Finally, the granted lifespan of patents was typically shorter than the requested term across most sectors. The average requested patent term per sector ranged from 8 to 13 years, while the granted term was usually 1 to 2 years shorter. This pattern is visible in major sectors such as textiles, transportation, food, and machinery, suggesting that the administration systematically adjusted patent terms when granting protection rather than simply approving the conditions initially requested by applicants.

Table D1: Descriptive statistics by technology class

Technology class	Applications			Granted patents			Granted invention patents				Granted importation patents			
	Total	Invent.	Import.	Total	Invent.	Import.	Total	Fee red.	Term req.	Term grant.	Total	Fee red.	Term req.	Term grant.
Textiles	139	62	77	79	68%	48%	42	12%	9.9	8.3	37	11%	10.7	8.1
Food	126	67	59	73	61%	54%	41	37%	9.9	8.3	32	9%	10.2	7.8
Transportation	152	78	74	73	55%	41%	43	21%	11.8	9.7	30	10%	11.6	8.3
Construction	69	48	21	36	63%	29%	30	20%	10.7	9.5	6	17%	13.3	11.5
Commodities	70	49	21	31	43%	48%	21	52%	8.3	7.1	10	30%	10.9	8.1
Utility	73	44	29	30	45%	34%	20	40%	10.0	9.8	10	30%	11.8	10.6
Machinery	43	26	17	29	69%	65%	18	22%	10.0	9.4	11	9%	12.4	9.7
Power	81	45	36	28	36%	33%	16	44%	13.3	11.6	12	0%	12.0	10.2
Instruments	42	35	7	24	60%	43%	21	57%	9.8	8.6	3	33%	10.0	5.7
Paper	47	30	17	24	73%	12%	22	50%	8.4	7.7	2	0%	10.0	8.3
Agriculture	38	29	9	17	48%	33%	14	36%	9.6	8.2	3	0%	13.3	6.7
Metallurgy	33	22	11	16	68%	9%	15	20%	10.3	9.0	1	100%	10.0	10.0
Mining	22	18	4	14	78%	0%	14	29%	11.9	11.1	0	n/a	n/a	n/a
Apparel	33	21	12	11	43%	17%	9	78%	9.4	9.4	2	50%	5.0	4.4
Chemicals	20	12	8	9	42%	50%	5	60%	13.0	9.0	4	0%	13.8	8.8
Military	19	12	7	9	50%	43%	6	33%	10.0	9.2	3	0%	14.7	8.3
Health	25	15	10	7	47%	0%	7	57%	9.3	9.3	0	n/a	n/a	n/a
Unknown	2	1	1	0	n/a	n/a	0	n/a	n/a	n/a	0	n/a	n/a	n/a
Total	1,067	614	420	532	56%	40%	344	34%	10.3	9.0	166	12%	11.2	8.5
Steam-related technology	70	29	40	42	48%	40%	20	10%	11.8	10.8	21	5%	12.8	9.8

Notes: Technology classes derived from Billington and Hanna, ‘That’s classified!’ Total numbers of requested and granted patents exclude patents of importation. Term requested and granted expressed in years.

Figure D1: Technology-class rankings and grant rates for patents of invention and importation



Notes: Derived from Table D1. Technology classes are ranked by the numbers of requested and granted patents, shown in parentheses. Arrow thickness represents the grant rate. Tied categories are ordered to minimise unnecessary changes in rank.

E PATENT QUALITY MEASUREMENT

Gerard Doorman was a Dutch economic and technological historian best known for his pioneering work on patents and industrial innovation in the Netherlands. The core of his 1947 book is a full listing of granted Dutch patents, supplemented with technological commentaries and archival information where available.⁵¹ Doorman compiled this material from patent registries at the Ministry for Industry, as well as patent archives deposited at the Ministry of Education, the ledgers of which we used to compile our own dataset of unsuccessful patent applications.⁵²

As stipulated by patent law, clerks in the UKNL maintained a registry listing all patents granted from the introduction of the law in January 1817 onwards. These registries recorded the patentee's name and residence, key administrative dates (initial submission, grant, and publication), the patent title, patent fees, and other relevant information such as special conditions, transfers, and nullifications. Doorman reproduced these registries with a high degree of fidelity and supplemented them with technological-historical commentaries.⁵³

We used Doorman's publication as the core dataset around which additional archival material was integrated. Optical character recognition software was used to convert the printed text into machine-readable form, after which the data were cleaned and used as a 'spine' to which we appended information from the original registries omitted by Doorman, primarily relating to patent fees and administrative milestones such as the date of first application.⁵⁴ This process produced a dataset covering all patents granted in the UKNL between 1817 and 1830.⁵⁵

Doorman's commentaries provide an indication of patent value, as he often situates inventions within their technological context or traces their origins and subsequent influence. He also notes awards received by inventors and references to inventions in contemporary publications. We use these commentaries to construct a 'Doorman indicator' of *ex post* patent value, assigning a value of one to patents discussed positively by Doorman. We argue that appearing in Doorman's index serves as a proxy for the *ex post* technological significance of a patent. This approach follows the spirit of Nuvolari and Tartari's Woodcroft index.⁵⁶ Bennet Woodcroft, head of the British Patent

⁵¹Doorman, 'Het Nederlandsch octrooiwezen'.

⁵²The registries, including a replacement registry made by Doorman (see footnote below), can be found under NL-HaNA, 2.04.23.01 (Ministerie van Binnenlandse Zaken: Afdeling Nijverheid en voorgangers, 1817-1877), inventory no. 1236-1240 (Registers van verleende octrooien). For the ledgers of the Ministry of Education, see NL-HaNA, 2.04.01 (Ministerie van Binnenlandse Zaken, 1813-1848), inv. no. 4049-4051 (Indices 1815-1818); 4190-4196 (Indices op de verbalen 1818 april – 1824 maart); 4925-4936 (Indices op de verbalen 1824-1830).

⁵³All registries have survived, except for the period 1829–1839. Doorman reconstructed a substitute registry for these missing years using the Staatscourant and several surviving indexes in the archives. These reconstructed registers contain substantially less information; patent fees and non-granted applications are not available.

⁵⁴For the patent dossiers, see NL-HaNA, 2.04.01, inv. no. 4039-4942 (Uitvindingen); inv. no. 4055-4175 (verbalen: besluiten, minuten van uitgaande brieven en ingekomen stukken (1818 april – 1824 maart); inv. no. 4204-4545 (verbalen 1824 april – 1830.)

⁵⁵We then added information on patent applications that were never granted, drawn from ministerial ledgers recording communications and decisions on patent requests. These ledgers document both unsuccessful applications and the administrative process underlying successful and unsuccessful patents.

⁵⁶Nuvolari and Tartari, 'Bennet Woodcroft'.

Office after the 1852 patent reform, commissioned an index of references in the contemporary literature to earlier patents. Nuvolari and Tartari use this index as a proxy for patent value on the assumption that more valuable inventions are more frequently discussed in the literature.

Table E1 reports the regional distribution of granted patents according to the Doorman indicator of *ex post* patent value. The majority of patents in all regions fall into the low-value category. Among applicants from the northern provinces of the UKNL, 83 patents are classified as low value and 17 as high value. The share of high-value patents is somewhat lower among applicants from the southern provinces and among foreign applicants. Overall, the table indicates that most patents across all regions do not appear to have generated inventions that were later recognised by Doorman as technologically significant.

Although the proportion of high-value patents is somewhat larger among applicants from the northern provinces, most high-value patents originate from the southern provinces in absolute terms. Applicants from the southern provinces account for 38 high-value patents, compared with 17 from the northern provinces and six from foreign applicants. This reflects the much larger overall volume of patenting activity in the southern provinces, which contribute around 70 per cent of all patents in the dataset and 62 per cent of the high-value patents.

Table E2 reports the distribution of low- and high-value patents using the Doorman indicator across technology classes, distinguishing between patents of invention and patents of importation. Overall, most patents fall into the low-value category, with 449 low-value patents compared with 61 high-value patents. This pattern holds for both patents of invention and patents of importation, although patents of invention account for the majority in both groups.

Across technological fields, low-value patents are widely distributed, with the largest numbers appearing in textiles (68), food (66), and transportation (61), followed by construction (33) and machinery (26). High-value patents are much rarer and more unevenly distributed across sectors. The largest concentrations appear in transportation (12), textiles (11), and utility technologies (7). In several sectors – including mining, apparel, and military technologies – no high-value patents are observed.

The relative importance of patents of invention and importation also varies by value category. Among low-value patents, patents of invention dominate (309 compared with 140 importation patents), and they also slightly outnumber importation patents among high-value patents (35 versus 26). Nevertheless, importation patents account for a meaningful share of high-value patents in sectors such as textiles, chemicals, and utility technologies, suggesting that imported technologies could also generate inventions that later proved economically significant.

Additional insight is provided by the category of steam-related technologies, which we construct from patent application titles containing the term ‘steam’. This category cuts across the technological classification we use in the rest of our analysis and is intended to capture innovations associated with one of the key general-purpose technologies of the Industrial Revolution. The data show that steam-related inventions were largely concentrated among low-value patents (38), with only three patents classified as high-value. Most of these patents

Table E1: Descriptive statistics on *ex post* valuable patents by applicant region

	Low value	High value
<i>Northern UKNL</i>		
<i>N</i>	83	17
% North	83%	17%
% UKNL	18%	28%
<i>Southern UKNL</i>		
<i>N</i>	315	38
% South	89%	11%
% UKNL	70%	62%
<i>Foreigners</i>		
<i>N</i>	51	6
% Foreign	89%	11%
% UKNL	11%	10%

Notes: Includes both patents of invention and importation. High-value patents are defined as those with a Doorman indicator.

were patents of invention rather than importation. This pattern suggests that while steam technologies generated considerable inventive activity, relatively few individual patents stood out in Doorman's retrospective assessment as having broader technological significance.

Taken together, these patterns indicate that patenting activity was broadly distributed across sectors, but patents that later attracted historical or technological recognition were both relatively rare and concentrated in a limited number of technological fields. Even within steam technology, most patented inventions did not subsequently stand out in Doorman's retrospective assessment. A comparable concentration can also be observed across the region of applicants' domicile: although the share of high-value patents is somewhat higher among applicants from the northern provinces, high-value patents remain concentrated among applicants from the southern provinces in absolute terms, reflecting the much greater concentration of patenting activity there.

Table E2: Descriptive statistics on *ex post* valuable patents by technology class

Technology class	Low-value patents			High-value patents			All patents		
	Invent.	Import.	Total	Invent.	Import.	Total	Invent.	Import.	Total
Textiles	38	30	68	4	7	11	42	37	79
Food	37	29	66	4	3	7	41	32	73
Transportation	34	27	61	9	3	12	43	30	73
Construction	28	5	33	2	1	3	30	6	36
Commodities	19	9	28	2	1	3	21	10	31
Utility	18	5	23	2	5	7	20	10	30
Machinery	16	10	26	2	1	3	18	11	29
Power	15	10	25	1	2	3	16	12	28
Instruments	19	3	22	2	0	2	21	3	24
Paper	20	1	21	2	1	3	22	2	24
Agriculture	13	3	16	1	0	1	14	3	17
Metallurgy	13	1	14	2	0	2	15	1	16
Mining	14	0	14	0	0	0	14	0	14
Apparel	9	2	11	0	0	0	9	2	11
Military	6	3	9	0	0	0	6	3	9
Chemicals	4	2	6	1	2	3	5	4	9
Health	6	0	6	1	0	1	7	0	7
Total	309	140	449	35	26	61	344	166	510
Steam-related technology	20	18	38	0	3	3	20	21	41

Notes: High-value patents are defined as those with a Doorman indicator; technology classes are derived from Billington and Hanna, 'That's classified!'; steam-related technologies are inventions related to steam power (see definitions in Table 1).

F A CONCEPTUAL MODEL OF DISCRETIONARY PATENT ADMINISTRATION

This appendix develops a conceptual model of the UKNL patent system, reconstructed from the archival evidence presented in Appendix A and the empirical findings of the main text. The equations formalise the mechanisms identified in our analysis and clarify the administrative decision problem. The model represents the challenge faced by UKNL officials: how to encourage useful invention and technology transfer while limiting unnecessary monopoly protection and facilitating access for applicants of limited means.

The model builds on the industrial organisation literature on patent design.⁵⁷ Three contributions are particularly relevant because they address the central elements of our model: the appropriate strength of patent protection, its adaptation to different economic circumstances, and the differentiation of patent rights across heterogeneous applicants. Gilbert and Shapiro distinguish between the size of the reward offered to innovators and its structure.⁵⁸ For a given reward, they analyse the trade-off between patent length and breadth. Klemperer similarly examines length and width, defined as the range of products covered by the patent, showing that the optimal combination depends on patterns of consumer substitution.⁵⁹ Cornelli and Schankerman introduce asymmetric information and differences in research productivity, showing that differentiated patent rights implemented through renewal decisions and fees can outperform a uniform patent term.⁶⁰ These models share the principle that patent protection should provide the necessary incentive at the lowest social cost.

Our model applies the same principle to a different problem. Rather than specifying *ex ante* how a legislature designs patent rights, we reconstruct how a bureaucracy allocated protection under incomplete information. The central question is administrative decision-making rather than optimal patent design. Using the information available to them, UKNL officials selected a package comprising the grant decision, fee, term, and restrictions on exclusivity, while considering applicants' financial constraints.

The model is best understood as an analytic narrative, using formal reasoning to clarify the administrative logic revealed by the archival evidence rather than deriving a general solution to optimal patent design.⁶¹ Accordingly, we do not specify functional forms, derive closed-form solutions, calibrate parameters, or estimate a structural model. Such exercises would require information on preferences, technologies, and expectations that is unavailable in the historical record. Instead, the model identifies the constraints and trade-offs implied by the archival evidence, and interprets the observed administrative decisions as consistent responses.

⁵⁷See Scotchmer, 'Innovation and incentives', for a textbook treatment of this literature.

⁵⁸Gilbert and Shapiro, 'Optimal patent length'.

⁵⁹Klemperer, 'How broad?'

⁶⁰Cornelli and Schankerman, 'Patent renewals'.

⁶¹See Bates et al., 'Analytic narratives', for the classic statement of the analytic narrative approach.

F.1 Applications and administrative information

Applicant i seeks either a patent for a novel invention or a patent of importation. Patent type and the information available to officials are represented as:

$$\widehat{v}_i^\tau = \mathbb{E}(v_i^\tau \mid z_i), \quad \tau_i \in \{N, M\}, \quad z_i = f(S_i, A_i, H_i). \quad (1)$$

where v_i^τ is the project's underlying gross private commercial value to the applicant, before implementation costs, patent fees, and limitations on appropriation. It depends on factors including novelty, practical usefulness, technical feasibility, and the economic circumstances in which the technology would be introduced. Officials cannot observe this value directly, so $\widehat{v}_i^\tau$ denotes their estimate based on the available information. The patent type is denoted by τ_i , where $\tau_i = N$ indicates a patent for a novel invention and $\tau_i = M$ a patent of importation.

The signal z_i summarises the information available to the administration when assessing a technology. It is generated by the information function $f(\cdot)$, which combines the patent specification S_i , reports from technical or industrial advisers A_i , and information gathered through ministerial investigation H_i . Officials use this information to update their assessment of the technology's expected value.

Implementing the technology entails a private cost, c_i^τ . For patents of novel invention, this cost also reflects the uncertainty associated with developing an unproven technology. We therefore assume that developing a novel invention generally entails at least as great an implementation cost as introducing a technology already proven abroad ($c_i^N \geq c_i^M$). The applicant's wealth, w_i , informs the fee selected by the administration. Implementation costs enter the participation decision directly, while wealth affects it through the fee. Officials observe both only imperfectly. Examination is therefore an information-producing process: corrected specifications, expert reports and correspondence improve the information available through S_i , A_i , and H_i , allowing administrators to update $\widehat{v}_i^\tau$ before reaching a decision.

The model therefore separates patent administration into two stages. Officials first estimate the technology's expected value from available information, then decide whether and on what terms to grant protection by weighing that estimate against broader social objectives.

F.2 Patent design and applicant participation

Rather than issuing a standardised patent right once the formal requirements were satisfied, the patent bureaucracy exercised substantive discretion over both the grant decision and the design of the resulting property right. It selected a patent package:

$$x_i = (g_i, F_i, T_i, \lambda_i), \quad P_i = T_i(1 - \lambda_i), \quad (2)$$

where x_i denotes the patent package, $g_i \in \{0, 1\}$ the grant decision, F_i the fee, T_i the term, and $\lambda_i \in [0, 1]$ any additional restriction, including stronger working requirements or compulsory

licensing provisions.⁶² The resulting level of patent protection is summarised by P_i , which increases with longer terms and decreases with stronger conditions. Administrators could therefore adjust the intensity of protection.

Let $\pi_i^\tau(P_i)$ denote the applicant's expected private return under patent protection. We assume that stronger protection increases this return, so that $\pi_i^{\tau'}(P_i) > 0$. The applicant proceeds when the expected return covers implementation costs and the patent fee:

$$\pi_i^\tau(P_i) - c_i^\tau - F_i \geq 0. \quad (3)$$

For any fee F_i , let $P_{i,\min}(F_i)$ denote the minimum protection required to satisfy this condition, and let P_i^* denote the protection selected by the administration. Conditional on the fee, any protection above this minimum would impose additional public costs without affecting participation. The administration therefore chooses:

$$P_i^* = P_{i,\min}(F_i). \quad (4)$$

Where positive protection is required, the participation condition binds: the applicant's expected return is just sufficient to induce development or introduction.⁶³

This conditional calibration is the model's central mechanism. The administration does not maximise the strength of the right granted but provides no more protection than is required given the fee it has selected. A lower fee reduces the protection required to induce participation, while a higher fee may require stronger protection. The fee choice determines which point on the participation frontier is implemented and is shaped by the fiscal and distributional considerations discussed below. Administrators could therefore combine fee reductions, shorter terms, licensing provisions, and working requirements to encourage useful invention or technology transfer without granting unnecessary monopoly protection.

F.3 Administrative objectives and patent type

The administration evaluated all applications using the same decision-making framework, but patents of importation could generate an additional social benefit through technology transfer:

$$U_i^\tau = g_i [\beta \widehat{v}_i^\tau + \phi_i^\tau - D(P_i) + \rho F_i] - K_i. \quad (5)$$

where U_i^τ denotes the application's expected net value from the administration's perspective; $g_i \in \{0, 1\}$, indicates whether the patent is granted; $\beta \widehat{v}_i^\tau$ is the direct economic benefit attributed to the project on the basis of its estimated private commercial value; and ϕ_i^τ captures additional public benefits arising specifically from technology transfer. Let $D(P_i)$ denote the public cost of

⁶²Patent fees were paid upfront at the time of grant rather than through subsequent renewal payments.

⁶³This reflects the administrative view, discussed in Appendix A, that patent holders should enjoy 'the greatest benefits' from their inventions, but only benefits that would serve them 'sufficiently as a reward', with patent duration 'limited to the shortest possible term' needed to induce development or introduction.

monopoly protection, with $D'(P_i) > 0$, meaning that stronger protection increases costs to the public by restricting competition and access to the protected technology. The term ρF_i captures the public value of fee revenue, while K_i is the administrative cost of examination.

We represent officials as treating their estimate of an applicant's private commercial value as an indicator of the project's broader social usefulness, including its potential to expand domestic production, introduce useful goods or processes, and contribute to industrial development. The parameter $\beta > 0$ captures the weight placed on this indicator, so that $\beta \widehat{v}_i^\tau$ represents the direct public benefit attributed to the project by the administration.⁶⁴

The term ϕ_i^τ captures the additional social value associated with technology transfer that is not reflected in the applicant's private commercial value, including benefits such as knowledge diffusion, learning through adaptation, and the acquisition of foreign technical capabilities. For patents of novel invention ($\tau_i = N$), $\phi_i^\tau = 0$, while for patents of importation ($\tau_i = M$), $\phi_i^\tau \geq 0$. It is strictly positive only where granting patent protection induces technology transfer that would otherwise not have occurred.⁶⁵

Fees created a trade-off: revenue was paid into a dedicated fund that supported patent administration and financed broader industrial-policy measures, but higher fees could discourage the development or introduction of technologies, especially by applicants of limited means. Administrators could therefore reduce the fee below the statutory standard rate:

$$0 \leq F_i \leq \overline{F}, \quad \frac{\partial F_i}{\partial w_i} \geq 0. \quad (6)$$

where $\overline{F}$ is the statutory standard fee and the maximum that could be charged, while F_i is the fee selected by the administration. The applicant's wealth w_i did not alter the statutory upper bound but informed the administration's choice within it. Because officials did not observe wealth directly, they used information gathered through ministerial investigation H_i to judge whether charging the statutory fee would prevent development or introduction. Financial and technical information served distinct purposes: inferred fee-paying capacity informed the fee, while technical examination informed the project's estimated private commercial value, $\widehat{v}_i^\tau$.

Both patent types are evaluated within the same framework: the administration first assesses the technology's expected value, then weighs it against technology-transfer externalities, monopoly and administrative costs, and fee revenue. Building on the assumption introduced in Section F1 that $c_i^N \geq c_i^M$, lower implementation costs for patents of importation may reduce the minimum protection required to induce participation. Holding the fee and return to protection constant, this implies:

$$P_{i,\min}^M(F_i) \leq P_{i,\min}^N(F_i). \quad (7)$$

⁶⁴The term $\beta \widehat{v}_i^\tau$ excludes the additional technology-transfer benefits captured by ϕ_i^τ and the monopoly costs captured by $D(P_i)$.

⁶⁵Without the term ϕ_i^τ , the model would distinguish the two patent types only through their private commercial value and implementation costs, omitting the public benefits of introducing foreign technologies.

Our empirical findings suggest that importers were likely wealthier, adding a countervailing effect: higher fees could increase the protection required and offset lower implementation costs. Grant probability also depends on whether protection was needed at all to induce technology transfer. If transfer would occur without protection, then $\phi_i^M = 0$ and monopoly costs weigh against granting a patent; technologies requiring substantial capital, adaptation, or specialist knowledge may still require protection. Patent type therefore has an ambiguous effect on grant probability, although conditions may differ systematically.

Outside advisers helped officials assess novelty, feasibility, domestic capacity, and effects on existing producers. Consultation could improve decisions but prolong examination, so adviser involvement need not systematically increase grant probability.

F.4 Discretion, discrimination, and empirical implications

The model distinguishes proportional treatment from discrimination. Differences in outcomes are proportional when they reflect expected value, implementation costs, ability to pay, or the protection required to induce technology transfer. Discrimination occurs when otherwise comparable applications receive different outcomes because of unrelated characteristics such as an applicant's nationality, region of origin, or social class.

A uniform regime assigns the same package $\bar{x}$ to every successful application, whereas a discretionary regime designs a tailored package x_i using the information in z_i . Let $\mathcal{W}^D$ and $\mathcal{W}^U$ denote aggregate welfare under discretionary and uniform regimes, respectively. Let Γ denote the benefit from tailoring, Ω its additional administrative cost, and Ψ the expected cost of inconsistency, favouritism, or opportunism. The net advantage of discretion is:

$$\Delta\mathcal{W} = \mathcal{W}^D - \mathcal{W}^U = \Gamma - \Omega - \Psi, \quad \Delta\mathcal{W} > 0 \text{ if } \Gamma > \Omega + \Psi. \quad (8)$$

Variation among applications and informative signals increase the benefits of tailoring; weak capacity, inconsistency, and bias increase its costs.

Implication 1: Case-by-case calibration

Where applications differ in expected value, implementation costs, applicant wealth and required incentives, discretion can outperform a uniform regime if administrative signals are informative and the costs of administration and bias are limited. Administrators then provide only the protection needed to induce implementation and reduce fees where limited means would otherwise prevent a useful project from proceeding.

A uniform package will provide excessive protection to some applicants and insufficient protection to others. Discretion allows adjustment to individual circumstances, but only when officials possess reliable information and apply consistent criteria. Even a well-designed discretionary system may otherwise perform poorly, while case-by-case variation may resemble

discrimination unless decisions follow intelligible principles. Our empirical analysis therefore examines whether outcomes responded to relevant application characteristics or instead reflected unexplained geographical or social differences.

The framework suggests fee reductions for applicants of limited means; adviser involvement and longer assessment for uncertain or complex applications; and importation terms reflecting whether protection was needed for technology transfer. Proportional discretion should produce variation along these dimensions without systematic residual differences by geographical or social origin.

F.5 Extension: Institutional feedback and divergence after 1830

The *de facto* dissolution of the UKNL in 1830 motivates an extension of the model.⁶⁶ Belgium and the Netherlands implemented their shared 1817 patent law in increasingly different ways. Belgium codified and invested in the system before replacing administrative discretion with more general rules in 1854, whereas the Netherlands neglected the system until its abolition in 1869.

Let O_{jt} denote patent requests (*octrooiaanvragen*) in jurisdiction j at time t , Y_{jt} industrial activity, and Q_{jt} the quality, accessibility and credibility of patent administration. Demand for patents is:

$$O_{jt} = f(Y_{jt}, Q_{jt}), \quad (9)$$

where $\partial O / \partial Y > 0$ and $\partial O / \partial Q > 0$, so patent demand is increasing in both industrial activity and the quality of patent administration. Let S_{jt} represent state investment in the patent administration. Institutional quality then evolves according to:

$$Q_{j,t+1} = (1 - \delta)Q_{jt} + \alpha S_{jt}, \quad S_{jt} = s(O_{jt}), \quad s'(O_{jt}) > 0. \quad (10)$$

Here $\alpha > 0$ measures the effectiveness of investment, and $\delta \in (0, 1)$ is the rate at which institutional quality deteriorates. Together, these relationships describe a self-reinforcing mechanism capable of generating virtuous or vicious circles. A well-functioning patent administration encourages patenting and investment, reinforcing institutional quality. Conversely, a poorly functioning system discourages patenting, weakens support for investment and reform, and becomes progressively less attractive.

Belgium followed the first path. Its government appointed dedicated patent examiners, codified the treatment of patents of importation, permitted fee payment by instalments, reformed examination procedures, and published patent specifications, making the system more transparent and accessible. As industrialisation increased patent use, however, case-by-case administration became increasingly costly and controversial. Belgium therefore retained patent protection while replacing much administrative discretion with clearer general rules.

⁶⁶This dynamic extension additionally draws on Wagenaar, ‘The patent system’.

The Netherlands followed the opposite path, culminating in abolition. Working requirements ceased to be imposed, specifications remained unpublished, the industrial fund was discontinued, and increasingly narrow judicial interpretation weakened enforceability. A growing share of granted patents never came into legal force because applicants did not pay the required fee, reducing the use of the patent system and therefore the constituency that might otherwise have demanded reform.

Let $j \in \{B, N\}$, where B denotes newly-independent Kingdom of Belgium and N the rump Kingdom of the Netherlands. Let $\mathcal{V}_{jt}^D$ be the perceived value of a discretionary system, $\mathcal{V}_{jt}^R$ that of a regularised patent system, and $\mathcal{V}_{jt}^0 = 0$ that of no patent system.⁶⁷ Each state selects:

$$\mathcal{V}_{jt}^* = \max \left\{ \mathcal{V}_{jt}^D, \mathcal{V}_{jt}^R, \mathcal{V}_{jt}^0 \right\}, \quad \begin{array}{l} \mathcal{V}_{Bt}^R > \mathcal{V}_{Bt}^D > \mathcal{V}_{Bt}^0, \\ \mathcal{V}_{Nt}^0 > \mathcal{V}_{Nt}^D, \mathcal{V}_{Nt}^R. \end{array} \quad (11)$$

In Belgium, industrialisation increased the value of patent protection while making discretionary administration increasingly costly, favouring regularisation. In the Netherlands, declining use, weak enforcement, and limited investment reduced the perceived value of patent protection, making abolition politically easier than reform.

Implication 2: Institutional feedback and divergence

Where industrial activity, patent use, and institutional investment reinforce one another, increasing use of a discretionary system makes regularisation more likely. Where limited use instead leads to institutional neglect, declining administrative quality may further reduce demand and make abolition more likely than reform.

Our model suggests that the advantage of discretionary patent administration lies in its ability to calibrate protection to heterogeneous applicants and technologies. This can reduce unnecessary monopoly protection while preserving incentives for socially useful invention and technology transfer. However, these benefits depend on sufficient administrative capacity, informative signals, and safeguards against inconsistent or discriminatory decision-making. The same mechanisms imply that discretion may be advantageous at one stage of industrial development but become increasingly difficult to sustain as patent use expands or administrative quality deteriorates. Discretion therefore disappeared for different reasons: Belgium outgrew it, whereas the Netherlands allowed the wider system in which it operated to deteriorate.

⁶⁷The regularised regime differs from the uniform regime introduced in the previous subsection. The latter is a theoretical benchmark in which every successful application receives the same patent package. The regularised regime instead represents the historical Belgian system after 1854, which retained limited scope for tailoring patent rights while replacing much case-by-case administrative discretion with codified rules and procedures.

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